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REVIEW SUMMARY

CANCER MICROBIOME

The microbiome and human cancer

Gregory D. Sepich-Poore, Laurence Zitvogel, Ravid Straussman, Jeff Hasty, Jennifer A. Wargo, Rob Knight*

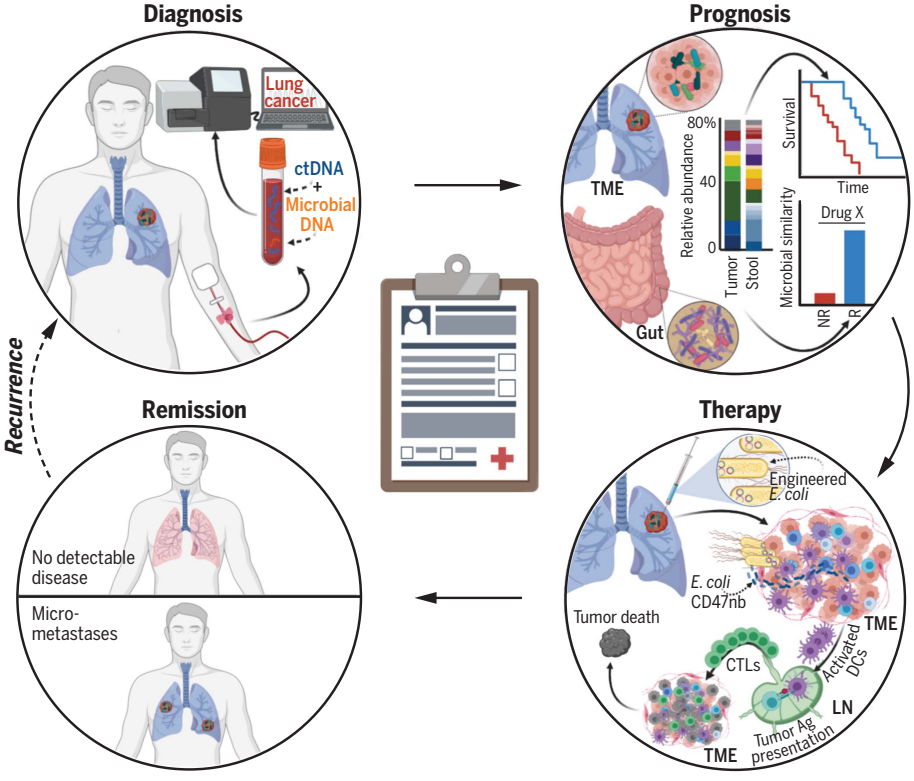
BACKGROUND: Historical accounts linking cancer and microbes date as early as four millennia ago. After establishment of the germ theory of infectious diseases, clinical research of microbial influences on cancer began in 1868, when William Busch reported spontaneous tumor regressions in patients with *Streptococcus pyogenes* infections. Over the next century, poor reproducibility, erroneous microbiological claims, and severe toxicity led many to discount the role of bacteria in carcinogenesis and cancer therapy. However, these studies provided the first crude demonstrations of cancer immunotherapy. Contemporaneously, the viral theory of cancer flourished, spurred by the 1911 discovery of Rous sarcoma virus, which transformed benign tissue into malignant tumors in chickens. The decades-long search to find viruses behind every human cancer ultimately failed, and many cancers have been linked to somatic mutations. Now the field is encountering intriguing claims of the importance of microbes, including bacteria and fungi, in cancer and cancer therapy. This Review critically evaluates this evidence in light of modern cancer biology and immunology, delineating roles for microbes in cancer by examining advances in proposed mechanisms, diagnostics, and modulation strategies.

ADVANCES: Few microbes directly cause cancer, but many seem complicit in its growth, often acting through the host's immune system; conversely, several have immunostimulatory properties. Mechanistic analyses of gut microbiota–immune system interactions reveal powerful effects on antitumor immunity by modulating primary and secondary lymphoid tissue activities. Many of these pathways invoke Toll-like receptor–initiated cytokine signaling, but microbial metabolic effects and antigenic mimicry with cancer cells are also important. In preclinical models, microbial metabolites also regulate phenotypes of tumor somatic mutations and modulate immune checkpoint inhibitor efficacy.

Emerging evidence suggests that intratumoral bacteria exist and are active, with overlapping immunohistochemistry, immunofluorescence, electron microscopy, and sequencing data in ~10 cancer types. Preliminary studies further suggest that fungi and bacteriophages contribute to gastrointestinal cancers. However, the abundance of intratumoral microbial cells is low relative to cancer cells, and knowledge of their functional repertoire and potency remains limited. Further validation of their prevalence and impact is needed in diverse cohorts and therapeutic contexts.

The immunomodulatory effects of host microbiota have reinvigorated efforts to change their composition as a form of immunotherapy. Despite extensive preclinical evidence, translation of microbiota modulation approaches into humans has not yet materialized into commercialized therapies. Synthetic biology approaches are also gaining traction, with engineered bacterial cancer therapies in preclinical and clinical trial settings.

OUTLOOK: A better understanding of the roles of microbes in cancer provides an opportunity to improve each stage of the cancer care cycle, but major challenges remain. Concerted efforts to characterize cancer-associated microbiota among tumor, stool, and blood samples with gold-standard contamination controls would tremendously aid this progress. This would be analogous to The Cancer Genome Atlas's role in characterizing the cancer somatic mutation landscape. Large-scale clinical trials are currently testing the efficacy of microbiota modulation approaches, ranging from dietary modifications to intratumorally injected, engineered bacteria. These bacterial cancer therapies, if safe and effective, could tremendously expand the cancer therapy armamentarium. Altogether, integrating the host-centric and microbial viewpoints of cancer may improve patient outcomes while providing a nuanced understanding of cancer-host-microbial evolution. ■



Opportunities for microbes to affect cancer care. Diagnosis: Cancer-specific, blood-borne microbial DNA may complement cell-free tumor DNA (ctDNA). Prognosis: Gut and intratumoral microbiota may stratify patient outcomes (NR, nonresponder; R, responder; TME, tumor microenvironment). Therapy: Intratumor injection of CD47 nanobody (CD47nb)–producing *Escherichia coli* may create systemic antitumor immunity by enhancing dendritic cell (DC) phagocytosis, lymph node (LN) antigen (Ag) presentation, and cytotoxic T lymphocyte (CTL) activity.

The list of author affiliations is available in the full article online.
*Corresponding author. Email: robknight@eng.ucsd.edu
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REVIEW

CANCER MICROBIOME

The microbiome and human cancer

Gregory D. Sepich-Poore[1], Laurence Zitvogel[2,3,4,5], Ravid Straussman[6], Jeff Hasty[1,7,8], Jennifer A. Wargo[9,10], Rob Knight[1,11,12]*

Microbial roles in cancer formation, diagnosis, prognosis, and treatment have been disputed for centuries. Recent studies have provocatively claimed that bacteria, viruses, and/or fungi are pervasive among cancers, key actors in cancer immunotherapy, and engineerable to treat metastases. Despite these findings, the number of microbes known to directly cause carcinogenesis remains small. Critically evaluating and building frameworks for such evidence in light of modern cancer biology is an important task. In this Review, we delineate between causal and complicit roles of microbes in cancer and trace common themes of their influence through the host's immune system, herein defined as the immuno-oncology-microbiome axis. We further review evidence for intratumoral microbes and approaches that manipulate the host's gut or tumor microbiome while projecting the next phase of experimental discovery.

The histories of cancer and human microbiota are intimately interwoven. Writings as early as 1550 BCE in the Ebers Papyrus, attributed to the Egyptian physician Imhotep (around 2600 BCE), suggest a crude treatment for tumors (swellings) involving application of a poultice to the site followed by an incision, causing an infection (*1*, *2*). In the 13th century, Peregrine Laziosi described spontaneous regression of his septic, ulcerative tibial bone malignancy that would have required amputation (*2*), for which he was canonized in 1726. After establishment of the germ theory of infectious disease, Wilhelm Busch and Friedrich Fehleisen independently reported in the late 1800s that *Streptococcus pyogenes* infections were associated with spontaneous tumor regressions in several patients (*3*, *4*). Shortly thereafter, William Coley started testing a highly contentious and sometimes lethal vaccine of live or heat-killed *Streptococcus* and *Serratia* species on terminal cancer patients, which was only later shown to yield >10-year disease-free survival in ~30% of them (60 of 210 total), representing the first intentional demonstration of immunotherapy (*5*). Contemporaneously, Thomas Glover and Virginia Livingston-Wheeler claimed, controversially, that bacteria were cultivable from tumors and that bacterial vaccines were effective against tumors, and suggested a universal bacterial origin of cancer (*6*, *7*). These early treatment approaches and theories were fraught with error: Livingston-Wheeler's bacterial "cause" of cancer, *Progenitor cryptocides*, turned out to be the skin commensal *Staphylococcus epidermidis* (a frequent contaminant), and Glover's findings were not reproducible by researchers at the National Institutes of Health (*7*). With no mechanistic evidence, irreproducible results, and hazardous therapies, the bacterial theory of cancer was dismissed.

The viral theory of cancer gained traction after Peyton Rous's 1911 discovery of a transmissible oncogenic virus in chickens (*8*). The subsequent decades-long search to find a virus behind every cancer linked Epstein-Barr, human papilloma, and hepatitis viruses to carcinogenesis (*9*) but failed to find a viral cause for most human cancers, and the theory was overtaken by the somatic mutation hypothesis.

Now, after decades of research thoughtfully characterizing the hallmarks of human cancer through somatic mutations and other host-centric perspectives (*10*, *11*), the field is encountering nuanced claims that microbes may play a broad role in cancer diagnosis, pathogenesis, and treatment (*12*–*26*). This reappraisal stems from greater appreciation of the number of microbes that inhabit the human body (roughly equal to the number of human cells), their gene count that exceeds the human genome's gene count by ~100-fold and enables diverse metabolic programming, and their effects on host immune system development and activity, including antitumor immunosurveillance (*27*–*31*). Although most proposed cancer-microbe relationships focus on gut microbiota (*30*, *32*, *33*), recent studies also contentiously suggest the existence, metabolic activity, and functional importance of intratumoral microbiota using a combination of imaging, sequencing, and cultivation techniques and genetically engineered and germ-free mouse models (*12*–*14*, *18*–*20*, *23*, *34*). These studies raise many questions about microbes and cancer. How should microbes be viewed in light of known host-centric cancer characteristics? To what extent are microbes causal agents, complicit actors, or passive bystanders? If intratumoral microbes exist, do they have therapeutic implications? What role do microbes play in patient management? With these questions in mind, this Review aims to critically evaluate the known roles of microbes in cancer and to outline the next steps for evaluating their clinical utility.

Overview of the cancer microbiome

Of the estimated ~10^{12} distinct microbial species on Earth (*35*), just 11 are labeled human carcinogens, or "oncomicrobes," by the International Association for Cancer Registries (IACR) (*36*). These oncomicrobes cause an estimated 2.2 million cases per year (~13% of global cancer cases), and their epidemiology, molecular mechanisms, and clinical studies have been extensively reviewed (*36*). Strong experimental evidence suggests that additional microbes initiate cancer through genotoxin-mediated mutagenesis; in particular, colibactin (a DNA alkylator), cytolethal distending toxin [CDT; direct deoxyribonuclease (DNAse) activity], and *Bacteroides fragilis* toxin (Bft; reactive oxygen species (ROS) producer) cause mutational signatures found in colorectal, head and neck, and urinary tract cancers (*22*, *37*–*41*). Experimental evidence also implicates several microbes with virulence factors that amplify tumorigenesis through E-cadherin–Wnt–β-catenin signaling, including FadA from *Fusobacterium nucleatum* and AvrA from several *Salmonella* strains (*42*, *43*). A few dozen microbial species can thus directly cause cancer, based on current epidemiological and experimental evidence.

Increasing evidence suggests an important additional category of "complicit" microbes and microbial functions that promote carcinogenesis but are insufficient to cause cancer (*18*, *20*, *25*, *38*, *44*–*47*). This category encapsulates many immunomodulatory functions of microbiota and their bioactive metabolites in tumor development and may be linked to the immune system's role in solid tumorigenesis; the immune system rarely initiates the incipient lesion but can facilitate progression through tumor-stroma feedback loops, inflammation, or dysfunctional immunosurveillance

[1]Department of Bioengineering, University of California, San Diego, La Jolla, CA, USA. [2]Gustave Roussy Cancer Campus (GRCC), Equipe Labellisée–Ligue Nationale contre le Cancer, Villejuif, France. [3]Institut National de la Santé et de la Recherche Medicale (INSERM) U1015, Villejuif, France. [4]Université Paris-Sud, Université Paris-Saclay, Gustave Roussy, Villejuif, France. [5]Center of Clinical Investigations in Biotherapies of Cancer (CICBT) 1428, Villejuif, France. [6]Department of Molecular Cell Biology, Weizmann Institute of Science, Rehovot, Israel. [7]BioCircuits Institute, University of California, San Diego, La Jolla, CA, USA. [8]Molecular Biology Section, Division of Biological Science, University of California, San Diego, La Jolla, CA, USA. [9]Department of Surgical Oncology, The University of Texas MD Anderson Cancer Center, Houston, TX, USA. [10]Department of Genomic Medicine, The University of Texas MD Anderson Cancer Center, Houston, TX, USA. [11]Department of Pediatrics, University of California, San Diego, La Jolla, CA, USA. [12]Department of Computer Science and Engineering, University of California, San Diego, La Jolla, CA, USA.
*Corresponding author. Email: robknight@eng.ucsd.edu

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(*11*). One example is that common p53 mutations are only carcinogenic in the presence of microbially produced gallic acid and are protective otherwise in the gut, both in vivo and in organoids, suggesting a microbiome-functional genomic interaction (*44*). A second is microbially produced secondary bile acids, which reduce hepatic sinusoidal CXCL16 expression (the sole ligand for CXCR6) and prevent $CXCR6^+$ natural killer T cell aggregation and liver cancer immunosurveillance—this carcinogenic effect is eliminated by vancomycin treatment (*45*). A third comes from the inability of Kras mutation and p53 loss to produce lung cancer in germ-free or antibiotic-treated mice: Commensal lung microbiota promote expansion and activation of γδ T cells, which drives tumor-promoting inflammation through local interleukin-17 (IL-17) and IL-23 release (*18*). These examples illustrate how microbes or microbial functions can be complicit in cancer rather than directly causal.

In contrast to the few bona fide oncomicrobes, the many complicit microbes and their functions are broad and understudied. Complicit microbes require mediators to promote tumor development but modulate tumor progression and therapeutic efficacy locally or from a distance. Complicit microbes are also least understood, requiring comprehension and integration of host and microbial biology, so we emphasize them in this Review. Together with known causal mechanisms, the diversity of these "complicit" mechanisms and their relationships to host-centric cancer hallmarks (*10*, *11*) are notable (Fig. 1), but they will require more rigorous experimentation and cross-cohort validation to establish clinical prevalence and utility.

Understanding the distribution of microbes across the body is important for understanding their relationships to cancer. About 4×10^{13} microbial cells spanning ~3×10^3 species inhabit the human body: About 97% of these cells are bacteria in the colon, ~2 to 3% are extracolonic bacteria (proximal gut, skin, lungs, etc.), and ~0.1 to 1% are archaea and eukarya (including fungi) (*27*, *48*). Human-infecting virus and phage counts and diversity may be greater (*49*). The high density of colonic bacteria is thought to drive most of the known microbial immunomodulatory effects in the mammalian intestinal tract, the largest immune organ in the body (*50*), but organ-specific commensals may exert their own overriding influence (*18*, *46*). Intratumorally, Nejman *et al.* used quantitative polymerase chain reaction (qPCR) of 16*S* ribosomal RNA (rRNA) to estimate the number of bacteria relative to 40 ng of DNA in melanoma, lung, ovarian, glioblastoma, pancreatic, bone, and breast cancer tissue sections (*12*). Assuming tissue homogeneity and 8 pg of DNA per cancer cell (based on 2.36 average tumor ploidy from the Pan-Cancer Analysis of Whole Genomes project) (*51*), the Nejman *et al.* data suggest an average pan-cancer percent bacteria composition at 0.68% bacterial (bootstrapped 95% confidence interval of mean: [0.52%, 0.87%], 1000 iterations), with individual tumors ranging from no bacteria to nearly 70% bacterial by cell count (*12*). Applying this percent bacterial composition to three-dimensional and planar contexts equates to ~10^5 to 10^6 bacteria per palpable 1-cm^3 tumor (*52*) or ~34 bacteria/mm^2 [assuming 5000 cells/mm^2 (*53*)], the latter of which is comparable to the average $PD1^+$ T lymphocyte tumor core density of ~21 cells/mm^2 from a recent pan-cancer cohort (*54*). Importantly, these bacterial composition estimates remain to be confirmed in other cohorts and cancer types and validated with orthogonal methods. Furthermore, which of these microbial taxa and functions can affect the host despite their low abundances remains unknown, as does the proportion that are merely passengers in a nutrient-rich and immunosuppressed space.

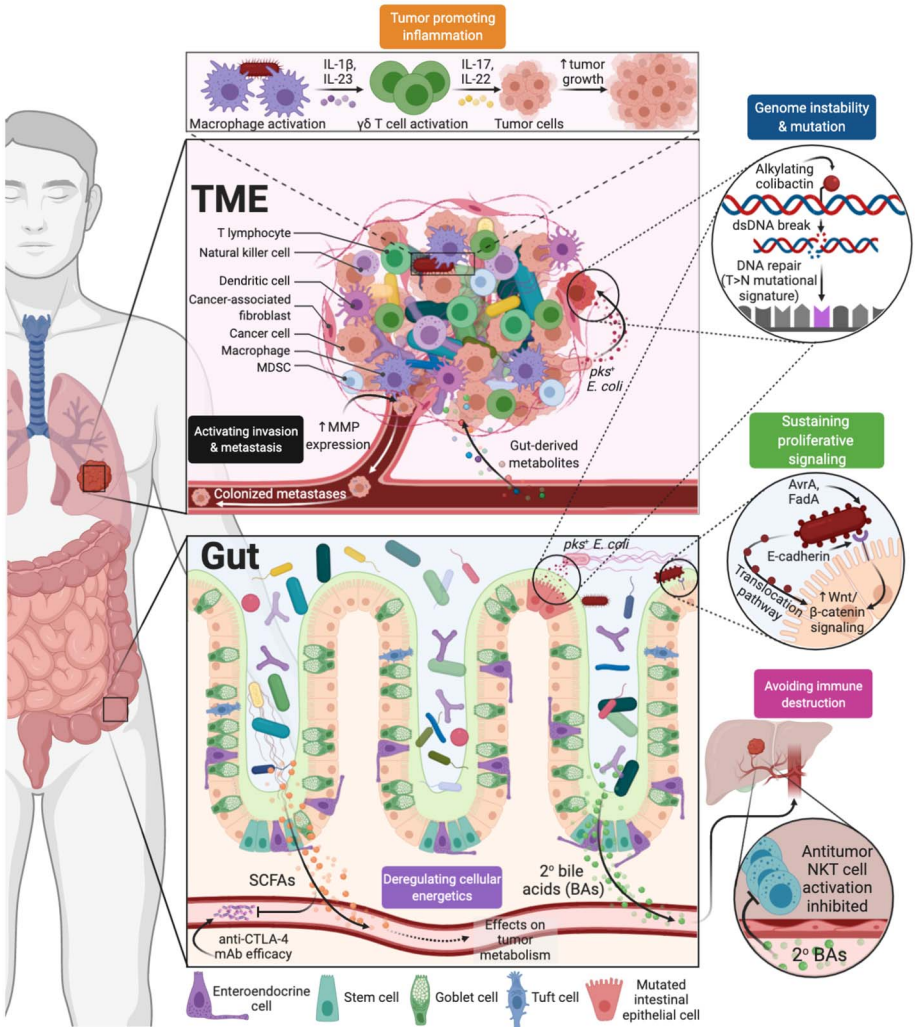


Fig. 1. Examples of the intersection of microbial mechanisms with cancer hallmarks. The cancer hallmarks are described in (*10*, *11*). Microbiota-derived metabolites, genotoxins, and antigens influence host antitumor immunity, inflammation, energetics, cellular signaling, and metastasis. dsDNA, double-stranded DNA; mAb, monoclonal antibody; MDSC, myeloid-derived suppressor cell; MMP, matrix metalloproteinase; NKT cell, natural killer T cell; SCFAs, short-chain fatty acids.

Mechanisms and interactions between the gut and tumor microbiome

Gut microbiota can regulate many functions of the tumor-bearing meta-organism, typically through immunomodulation, and putative intratumoral microbes may also be important (*31*, *55*). Known microbial mechanisms can manipulate nonhematopoietic and hematopoietic components of the gut epithelial barrier, modulate primary and secondary lymphoid organ activities, and regulate immune tone of the tumor microenvironment (TME). We define these immune-mediated interactions

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and collective feedback loops as the immuno-oncology-microbiome (IOM) axis (Fig. 2). Gut-TME cross-talk, especially in nongastrointestinal cancers, remains a key area of discovery.

The effects of gut microbiota on primary lymphoid organs

After allogeneic hematopoietic stem cell transplantation (HSCT), robust immune reconstitution governs both relapse and transplant-related patient mortality (*56*, *57*). A recent multicenter, multinational clinical trial demonstrated that higher diversity of intestinal microbiota is significantly associated with lower patient mortality after allogeneic HSCT (*58*). Moreover, in an allogeneic HSCT trial that coanalyzed daily changes in patient differential blood counts with >10,000 longitudinal fecal samples, immune-reconstitution dynamics were closely linked to gut microbiota composition (*59*). Links between gut microbiota, nutrition, post-transplant bone marrow (BM) and thymic cellularity, and lympho- and myelopoiesis have also been demonstrated in mouse models (*57*). Gut microbe depletion impairs systemic infection clearance after BM transplant and sensitizes mice to semilethal doses of radiation. Microbiota-derived compounds can protect against irradiation-induced hematopoietic injury (*60*–*62*) through production of propionate and tryptophan metabolites (*63*) or by releasing microorganism-associated molecular patterns (MAMPs) known to maintain BM-derived myeloid cells and neutrophil function (*64*, *65*). This effect may be explained in part by the delivery of endogenous ligands for RIG-I (such as 3pRNA and RNA derived from viruses, phages, or bacteria) that can induce protective type I interferon (IFN-I) signaling in enterocytes and intestinal barrier repair (*66*). Post-transplant lymphopoiesis also depends on energy harvest from the diet and potentially on genera whose genomic repertoires encode carbohydrate-active enzymes (*57*).

The effects of gut microbiota on adaptive immunity

The gut microbiota has broad effects that contribute to host immune tone at steady state and during tumorigenesis (*30*, *67*). Anticancer therapies have demonstrated strong links between distinct commensals and protective antitumor T cell responses: (i) Cyclophosphamide enables *Enterococcus hirae* to translocate and stimulate pathogenic helper T cell 17 (T_H17) responses and IFN-producing $CD8^+$ T cell effectors that check tumor growth in sarcoma and lung adenocarcinoma models (*68*, *69*); (ii) in some patients with melanoma, cytotoxic T lymphocyte–associated antigen 4 (CTLA-4) blockade allows fecal relative enrichment of *Bacteroides thetaiotaomicron* and *B. fragilis*

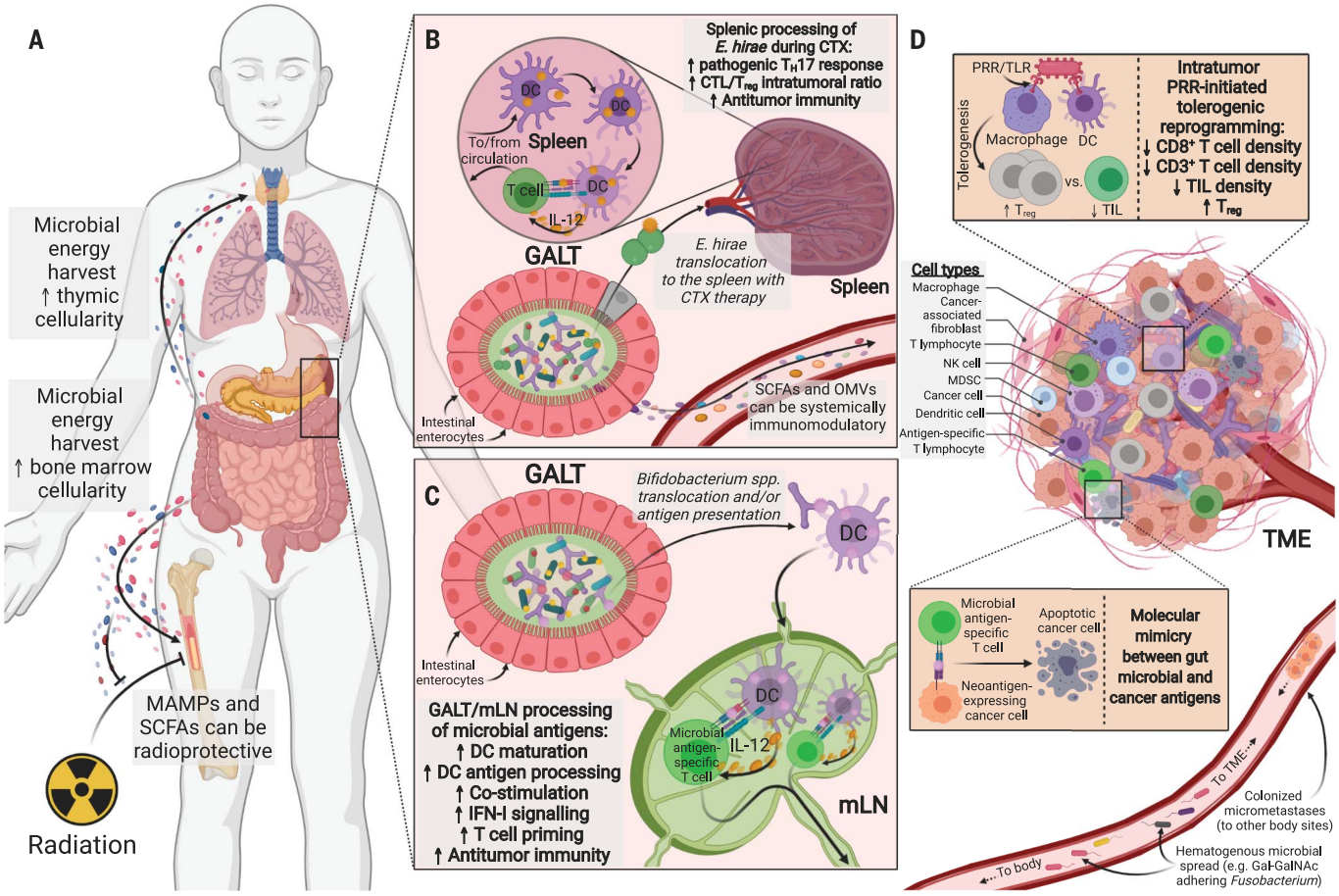


Fig. 2. Defining the IOM axis. Gut and TME microbiota regulate host metabolism and immunity, which ultimately influence antitumor immunity. (**A**) Gut microbial metabolites and by-products influence host lympho- and myelopoiesis, including during allogeneic HSCT and radiotherapy (*59*, *63*). (**B**) Cyclophosphamide (CTX)–derived gut epithelial damage enables *E. hirae* translocation and antitumor immunity (*68*, *69*). (**C**) Gut translocation of *Bifidobacterium* species or its antigens can increase IFN-I signaling and antitumor immunity (*73*, *76*). (**D**) Microbes within the TME can be either immunosuppressive (often PRR-mediated) or immunogenic, including shaping the response to immunotherapy (*12*, *23*). Cancer (neo)antigens may share epitopes with microbes through molecular mimicry (*73*, *83*). Microbial hematogenous spread (*117*, *172*, *173*) or colonized micrometastases (*19*) may complete this feedback loop that originated in the gut. CTL, cytotoxic T lymphocyte; DC, dendritic cell; GALT, gut-associated lymphoid tissue; MAMPs, microbe-associated molecular patterns; mLN, mesenteric lymph node; NK cell, natural killer cell; OMVs, bacterial outer membrane vesicles; SCFAs, short-chain fatty acids; TIL, tumor-infiltrating lymphocyte.

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that mediates Toll-like receptor 4 (TLR4)– and IL-12–dependent T_H1 responses and therapeutic efficacy (*70*); (iii) inhibition of programmed cell death protein 1 and/or its ligand [PD-(L)1] leads to T cell priming against melanoma and is more effective when hosts harbor *Bifidobacteria* species in their microbiome (*21*, *71*); (iv) adoptive T cell transfer efficacy against melanoma after total body irradiation depends on the composition of the microbiota, the translocation of gut bacteria, and host TLR4 signaling (*72*–*74*); (v) oxaliplatin-induced cell death of ileal enterocytes inversely governs the immunogenic *Erysipelotrichaceae* and tolerogenic *Fusobacteriaceae* proportions in the ileum, dictating the balance between antitumor follicular T helper cells and deleterious T_H17 responses in colon cancer (*75*).

In most of these models, dendritic cells (DCs) from the gut-associated lymphoid tissue (GALT), spleen, or tumor draining lymph node (LN) sense various commensals (*Bifidobacterium* spp., *B. fragilis*, *Akkermansia muciniphila*, *Bacteroides rodentium*, *Bacteroidales* S24-7), catalyzing immune responses through IFN-I– and IL-12–mediated pathways (*17*, *70*–*72*, *75*–*77*). Apart from providing DC adjuvants, the gut microbiota represent an antigen source that can elicit commensal-specific T cell responses systemically (*55*, *78*). In the context of homologous self-antigens, these commensal-specific immune responses can be deleterious or protective for the host, depending on the involved peptide(s). For instance, Gil-Cruz and colleagues demonstrated how homology between *B. thetaiotaomicron*–derived β-galactosidase and host cardiac myosin heavy chain 6 could drive lethal autoimmune inflammatory cardiomyopathy (*79*); conversely, Nanjundappa and others reported how cross-reactivity between *Bacteroides* species–derived integrase and host islet-specific glucose-6-phosphatase catalytic subunit-related protein (IGRP) could hijack autoreactive $CD8^+$ T cells to instead suppress colitis (*80*). Recent studies have further expanded this cross-reactive homolog list to include exogenous dietary antigens, notably between gliadin epitopes and gut *Pseudomonas fluorescens*–derived succinyl-glutamate desuccinylase in the context of HLA-DQ2.5–mediated celiac disease (*81*). Molecular mimicry between cancer and microbial antigens has also been hinted (*82*) and recently studied in depth (*83*). H-2Kb–restricted T cell immune responses against a phage that infects distinct strains of enterococci (*E. hirae*) cross-reacted with an oncogenic driver (PSMB4). Oral administration of *E. hirae* strains containing this phage then boosted phage-specific T cell responses effective against extraintestinal tumors overexpressing *PSMB4* during therapy with cyclophosphamide or anti–PD-1 antibodies (*83*). Similarly, T cells targeting an epitope, SVYRYYGL (SVY), expressed in the commensal bacterium *Bifidobacterium breve*, cross-reacted with a model neoantigen, SIYRYYGL (SIY), expressed by mouse melanoma B16-SIY (*73*). Moreover, some human T cells specific for naturally processed melanoma epitopes were found to recognize microbial peptides (*83*), suggesting clinical importance. However, mechanisms outside of molecular mimicry that boost antitumor immunity must also exist. For example, Tanoue and colleagues identified an 11-bacteria cocktail that increased tumor antigen–specific $CD8^+$ IFN-γ^+ T cells in the context of immune checkpoint blockade that were not cross-reactive with microbial antigens and did not originate from the colon (*84*).

Gut-derived metabolites can also modulate immune responses. Radiotherapy of tumor lesions was more effective when vancomycin eliminated *Clostridiales*-derived immunosuppressive metabolites (butyrate and propionate) putatively by increasing DC antigen presentation and concomitant $CD8^+$ T cell priming (*72*); conversely, gut microbial–derived propionate and tryptophan pathway metabolites (^{1}H-indole-3-carboxaldehyde, kynurenic acid) were shown to provide long-term radioprotection in vivo (*63*). High blood butyrate and propionate levels were also associated with resistance to CTLA-4 blockade in mice and melanoma patients, with concomitantly increased regulatory T cell (T_{reg}) proportions, reduced DC and effector T cell activation, and lower responses to IL-2 (*85*), although they were also found to be associated with longer progression-free survival during anti–PD-1 treatment (*86*). Moreover, increased ex vivo outgrowth of *A. muciniphila* by prebiotic mucin decreased growth kinetics of aggressive melanoma in a gut microbiota– and T cell–dependent manner, reducing serum levels of proinflammatory and immunosuppressive IL-6, IL-1α, IL-10, IL-17A, and IL-23 cytokines (*87*); notably, prebiotic inulin operated through a different mode of action, facilitating the dominance of *Bifidobacteria* species in the intestines, boosting splenic cytotoxic T lymphocyte functions, and overcoming melanoma resistance to MEK inhibitors (*87*).

The effects of gut microbiota on the TME

The intestinal ecosystem can influence both local and distant neoplasia by affecting their immune context, influx of myeloid and lymphoid cells, and inflammatory and metabolic patterns. Secretory components of gut microbiota can be important: For example, outer membrane vesicles (OMVs) can reprogram the TME toward a pro-T_H1 pattern (CXCL10, IFN-γ) (*88*), or metabolites, including butyrate and niacin, can mediate Gpr109a-dependent induction of IL-18 in colonic epithelium and suppress colitis and colon cancer (*89*).

Tumor-associated, NOX2-mediated, myeloid cell ROS production is reduced by antibiotic administration or germ-free status, reducing oxaliplatin's capacity to mediate early tumor genotoxicity (*90*). Similarly, commensal microbiota primed tumor-associated innate myeloid cells for tumor necrosis factor–α (TNFα) (IL-1β, IL-12, and Cxcl10) production in response to anti–IL-10R/CpG-ODN treatment, and antibiotics, germ-free, or $TLR4^{-/-}$ status attenuated this response and the TNF-dependent early tumor necrosis (*90*). Supporting the adjuvant role of commensals against developing cancers, pasteurized *A. muciniphila* or its pili-like TLR2 agonist blunted azoxymethane-induced colitis and colon carcinogenesis by inducing TNFα-producing cytotoxic T lymphocytes in mesenteric lymph nodes (mLNs) that eventually reached the colonic mucosa (*91*). Spontaneous gut bacterial translocation in $Tet2^{-/-}$ mice also drove preleukemic myeloproliferation (PMP), which leads to leukemia if unchecked, in an IL-6–dependent manner (*47*). PMP was reversible with antibiotics and abolished in germ-free mice, suggesting new clinical management opportunities. However, an intact gut microbiome was later shown to be necessary to prevent leukemia progression in genetically predisposed mice (*92*).

Nonhematopoietic components of the intestinal mucosa are also linked to the TME (*77*). Gene-deficient mice and BM chimeras identified a role for RNF5, an E3 ubiquitin ligase, in immunosurveillance of severe melanoma. $Rnf5^{-/-}$ mice exhibited decreased secretion of antimicrobial peptides and increased cell death in the ileal crypts, causing changes in intestinal microbiota community composition. This bowel injury amplified mobilization of CCR7-expressing DCs to Peyer's patches, mLNs, and melanoma-draining LNs, increasing IFN-γ–producing T lymphocyte tumor infiltration. Confirming a $Rnf5^{-/-}$-specific microbial effect, cohousing $Rnf5^{-/-}$ mice with wild-type mice, or administering antibiotics, restored tumor aggressiveness, whereas oral gavage with 11 species overrepresented in $Rnf5^{-/-}$ animals (*Bacteroides* and *Parabacteroides* spp.) into germ-free wild-type mice recapitulated tumor immunosurveillance (*77*). In another study, oxaliplatin-induced caspase 3/7–dependent ileal apoptosis of crypts coincided with immunogenic bacteria dominance in the ileal mucosa (*75*). These commensals regulated follicular T helper cell priming in mLNs, culminating in B cell activation, immunoglobulin production, and infiltration of colon cancers with tumor-infiltrating lymphocytes (TILs) in mice and patients. Anti–CTLA-4–induced gut barrier dysfunction was also critical for systemic translocation of *Bifidobacterium*-derived inosine, in turn promoting T_H1 activation and antitumor immunity by agonizing T cell–specific adenosine 2A receptor ($A_{2A}R$) signaling in the context of DC costimulation (*93*). These examples illustrate that barrier injury is accompanied by

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a deviation of the local microbiome or translocation of microbial metabolites that, in turn, mobilizes DCs to and outside the GALT and contributes to tumor bed infiltration by activated helper or cytotoxic T cells.

The TME comprises not only stromal, tumor, endothelial cells, and hematopoietic progenitor–derived immune components but also a dense network of adrenergic nerve fibers that influence oncogenesis in brain and nonbrain cancers (*94*–*97*). Interestingly, enteric nervous system neurons are both affected by the gut microbiota and functionally tuned according to their location in the gut. A subset of microbiota-responsive neurons could influence metabolic control independent of the central nervous system (*98*). These findings suggest intimate relationships between mucosal or tumoral commensals and tumor innervation that need further study.

Gut microbiota–mediated effects on anticancer drugs

Gut microbes are intimately involved in the biotransformation of xenobiotics, including cancer drugs, with unintended consequences for clinical cancer control (*99*). For example, in prostate cancer, abiraterone acetate (AA) was used as an energy source by *A. muciniphila* and inhibited *Corynebacterium* species relying on AA-inhibited androgens for growth (*100*). Because *A. muciniphila* is anti-inflammatory and *Corynebacterium* species are proinflammatory, this change in their relative abundances increased the pharmacologic efficacy of AA therapy. *A. muciniphila*'s immunomodulatory effects (*78*), including association with responders during PD-1 blockade (*17*), has prompted speculation that increased *A. muciniphila* may explain the efficacy of AA in androgen-independent prostate cancer (*100*), although this remains to be tested in large patient cohorts. These types of bidirectional drug-microbiota feedback loops warrant further study.

Intratumor microbiota effects on the TME

Mechanistic studies of live microbiota within diverse tumor types have been limited, particularly outside the aerodigestive tract, but many of their effects on the TME appear to suppress local antitumor immunity (*15*, *23*, *34*, *46*, *75*, *101*, *102*). Additionally, intratumor microbes have been reported to have cancer-specific effects on (i) gastrointestinal and urinary tract mutagenesis through secreted genotoxins, most notably *pks*$^+$ *Escherichia coli*–derived colibactin and *B. fragilis*–derived toxin (*22*, *37*, *38*, *40*, *41*); (ii) CagA-mediated or IL-17–producing γδ T cell–mediated inflammation in stomach and lung cancers, respectively (*18*, *103*); (iii) chemoresistance through direct microbial metabolism (cytidine deaminase degradation of gemcitabine) in pancreatic cancer or indirect amplification of cancer cell autophagy in colorectal cancer (*14*, *104*); (iv) tumor proliferation through fungal activation of the host's C3 complement cascade in pancreatic cancer (*20*); and (v) metastasis through up-regulation of tumor matrix metalloproteinases in breast cancer or reduction of tumor immunosurveillance in lung cancer (*34*, *46*). Immunologically, intratumor microbes often create tolerogenic programming through pattern recognition receptor (PRR) ligation with lower proportions of TILs, including CD8$^+$ T cells, and occasionally more CD4$^+$CD25$^+$FoxP3$^+$ T_{regs}, as observed in colorectal, pancreatic, breast, and lung cancers (*18*, *23*, *34*, *46*, *75*, *101*, *102*). However, in certain cases, injection of intratumoral bacteria or their antigens may conversely provide immunostimulatory effects, as demonstrated by Coley's toxins and recent developments in bacterial cancer therapy (*5*, *105*, *106*). In breast cancer, experiments comparing SCID-beige and C57BL/6 mice with intratumor *Fusobacterium* suggested lymphoid-lineage cells as key mediators of intratumor microbiota-derived effects on tumor immunosurveillance (*34*). There are also associations between intratumor microbiota and immunogenicity, including differential melanoma immunotherapy response and triple-negative breast cancer associations, but their underlying mechanisms remain uncharacterized (*12*).

Extraintestinal barriers and cancer microbiota

Given that the intestinal barrier offers the largest host-microbial interface and greatest microbial diversity, investigations on the potential impact of the microbiota in oncogenesis or cancer prognosis have primarily focused on this barrier (*107*). These studies could unveil cause-effect relationships between gut microbial composition changes and compromised tumor immunosurveillance, even in extraintestinal malignancies. However, it is noteworthy that extraintestinal cancers can develop within tissues that harbor their own microbiome and may play a role in the exacerbation of neoplasia (*12*, *18*, *25*, *46*).

For instance, the lungs' surface approximates 1 m^2 per kg of body weight and is not sterile (*108*). Experimental evidence in oncogene-driven autochthonous lung cancer models in mice unveiled that local commensals may be perturbed by carcinogenesis, triggering an inflammatory cross-talk between alveolar macrophages and IL-17–producing lung resident γδ T cells contributing to tumor progression (*18*). The clinical importance of this observation has been recently brought up in 83 lung cancer patients (*25*). Tsay *et al.* highlighted that microaspiration of supraglottic commensals in lung cancer patients can affect response to therapies and overall survival, owing to a T_H17–mediated exacerbated inflammation corollary to immune checkpoint inhibition (*25*).

Skin is also recognized as our largest and outermost organ, maintaining host homeostasis through tight interconnections between its resident microbes, keratinocytes, and skin immune components through metabolic, innate, and cognate immune responses (*109*). Compositional shifts in the skin microbiota appear to influence nonmelanoma skin carcinogenesis (*110*). Similarly, cervical cancer caused by persistent high-risk human papillomavirus infection is often associated with a deviated cervical microflora (*111*, *112*). The intertwined and/or interkingdom relationships between commensals and virus-associated cancers and their synergistic effects on tumorigenesis need further study, and exploration of cancer-microbe interactions at other extraintestinal barriers is warranted.

Cancer microbiome diagnostics

Variation in human microbiome composition among body sites (*113*) contrasts with stable human genetics that exhibit only minor variation resulting from somatic mosaicism and clonal hematopoiesis (*114*). Because both host tissues and microbiota are affected by carcinogenesis, the genetic heterogeneity of microbes may provide an opportunity to diagnose and locate disease. For example, a blood-derived TP53 mutation can indicate host cancer status but implicates >25 cancer types (*115*); conversely, *Streptococcus gallolyticus* (formerly *S. bovis*) bacteremia can reflect host cancer status and type (colon cancer) based on its gastrointestinal origin (*116*, *117*). Many challenges exist for microbial-based diagnostics, including low biomass relative to host and confounding from reagent or environmental contaminants. Many questions about their specificity, prevalence, and stability during cancer treatment or utility during antibiotic administration remain to be answered and must be addressed before clinical deployment.

Nearly all microbial-based cancer diagnostics are sequencing-based and have focused on tumors within the aerodigestive tract (*31*), such as colorectal (*118*–*121*), pancreatic (*122*, *123*), and lung cancer (*124*–*126*). It was only recently suggested that cancer types outside of the aerodigestive tract, such as breast or brain cancer, may also harbor microbiota with distinctive compositions. Nejman *et al.* (*12*) suggest distinct intratumoral microbiomes among seven cancer types (Fig. 3), providing imaging evidence of these microbes' intratumoral spatial distribution and intracellular localization.

Combining multiregion 16*S* rRNA amplicon sequencing, qPCR, immunohistochemistry [lipopolysaccharide (LPS), lipoteichoic acid (LTA)], immunofluorescence (16*S* rRNA), cultivation, and electron microscopy, Nejman *et al.* (*12*) surveyed 1010 tumors for bacteria across melanoma, lung, ovarian, glioblastoma, pancreas, bone, and breast cancers. They included

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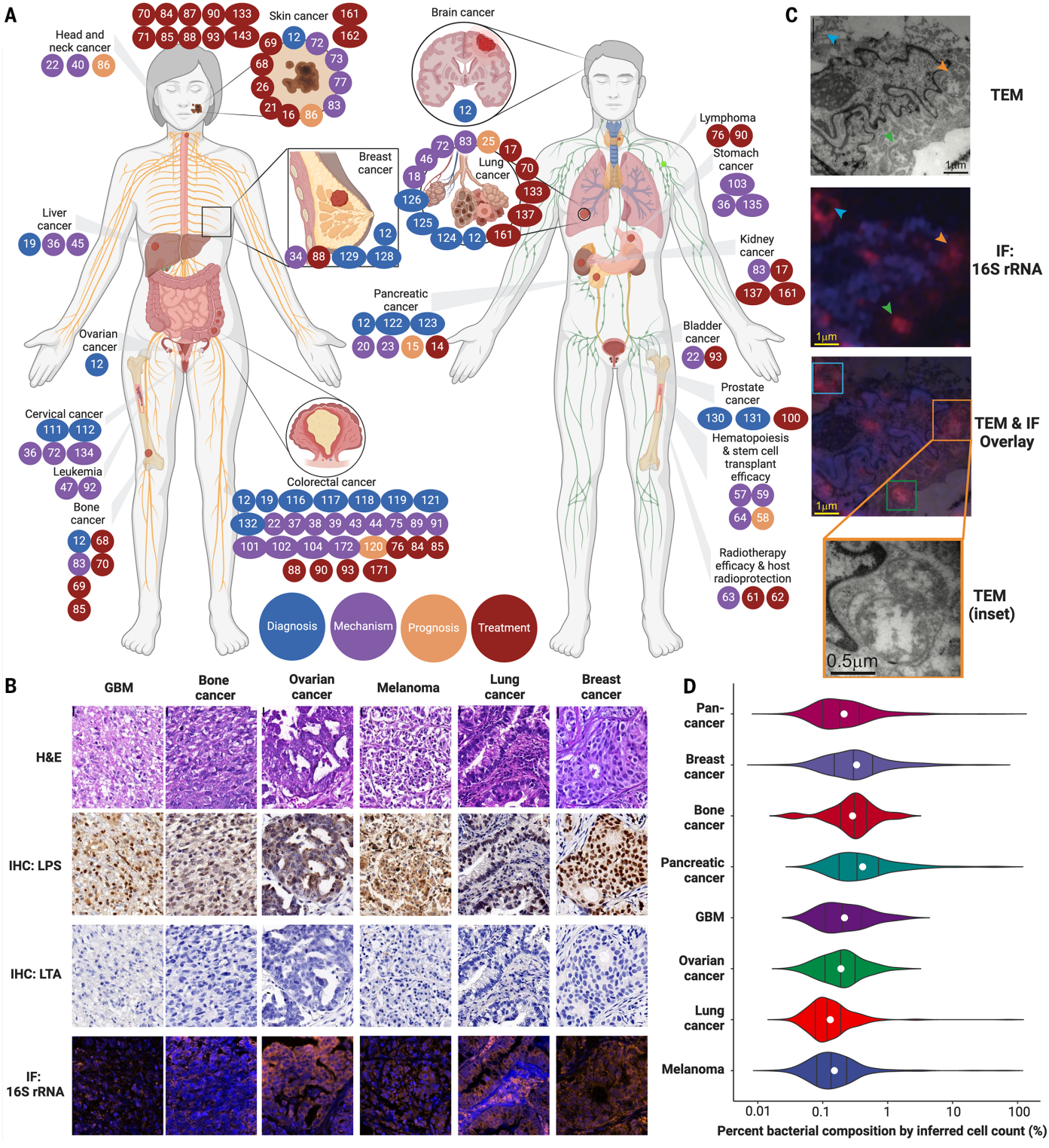


Fig. 3. Current landscape of the cancer microbiome. (**A**) Body diagram of all cancers currently linked to microbiota, where the colored dots reflect reference numbers and are colored according to the major theme of the referenced paper (diagnosis, mechanism, prognosis, or treatment). Dots are included based on existing preclinical and clinical data. (**B**) Representative histology, immunohistochemistry (IHC) for lipopolysaccharide (LPS) and lipoteichoic acid (LTA), and immunofluorescence (IF) for bacterial 16*S* rRNA in six cancers. H&E, hematoxylin and eosin; GBM, glioblastoma. (**C**) Representative transmission electron microscopy (TEM) images with overlaid 16*S* rRNA immunofluorescence of intracellular bacteria (arrows) in breast cancer. (**D**) Estimation of tumor percent bacterial composition across seven cancer types, assuming tissue homogeneity and 8 pg of DNA per cancer cell. Black lines depict distributional quantiles (25%-50%-75%); white dots reflect averages. Images in (B) to (D) are adapted from Nejman *et al*. (*12*).

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811 experimental controls, covering DNA extraction, PCR amplification, and paraffin embedding, which identified and removed 94.3% of bacteria as contaminants. Examining the residual 528 bacterial species revealed substantial differences in composition, diversity, and inferred metabolic functionality between cancer types. Histologic imaging revealed heterogeneous microbial spatial distributions (Fig. 3B) and their frequent intracellular localization in cancer and immune cells (Fig. 3C). As described above in the "Overview of the cancer microbiome" section, qPCR estimated the number of bacteria per tissue section, which we have graphically depicted as percent bacterial composition per cancer type, assuming tissue homogeneity and 8 pg of DNA per cancer cell (Fig. 3D). Applying their pipeline to a melanoma immunotherapy cohort suggested microbiome differences between responders and nonresponders, but not yet a mechanism. Because bacteria were cultured from only five human breast tumors, the widespread viability of intratumoral bacteria from this study was unclear, particularly in cancers with reportedly fewer bacteria. However, other studies have indeed shown cultivable bacteria in breast (*127*–*129*), lung (*18*), prostate (*130*, *131*), pancreas (*14*, *15*), and colon cancers (*19*, *132*), suggesting broad microbial viability. Still, basic questions remain about the functional impacts of these intratumoral microbiota and whether they are parasitic, symbiotic, or passive passengers, and a biopsy specimen is required for analysis, limiting its diagnostic utility. Moreover, although the intracellular bacteria in cancer and immune cells identified by Nejman *et al.* (*12*) may provide one source of blood-derived microbial DNA, this remains to be mechanistically demonstrated. A rigorous evaluation alongside blood samples from patients with nonlethal bodily infections, septic patients, and patients receiving antibiotics during cancer care are necessary preconditions to broad clinical utility.

Modulation of the cancer microbiome

The associations between certain gastrointestinal microbiota and the activity of systemic lymphoid tissues have stimulated interest in microbial modulation as a powerful immunotherapeutic modality. If intratumoral microbiota are eventually verified to be prevalent and immunologically active across many patients, as preliminary data suggest (*12*), such interventions must account for microbial niches and their cross-talk (Fig. 4). These dynamics sometimes appear related; for example, modulation of gut microbiota influences the composition of the intratumoral microbiome in pancreatic cancer, presumably through pancreatic duct communication (*15*, *20*, *23*). However, in other cases, these changes are incongruent; for instance, antibiotics appear to abrogate immunotherapy response by inhibiting the gut microbiome (*133*), but paradoxically they improve immunotherapy efficacy by upregulating PD-1 expression when eliminating the pancreatic intratumoral microbiome (*23*). These complexities necessitate more in-depth mechanistic studies of modulation approaches and better clinical understanding before applying prebiotics, probiotics, postbiotics, and antibiotics in the setting of cancer.

Antibiotics and the cancer microbiome

The use of antimicrobial therapy in cancer is limited to addressing or preventing known microbial carcinogens. This includes treating *Helicobacter pylori*–derived gastric lymphomas with triple or quadruple antibiotic therapy, administering direct-acting antivirals against active hepatitis C virus, and vaccinating against major human papillomavirus serotypes and hepatitis B virus to prevent urogenital, cervical, head and neck, and liver cancers (*36*, *134*, *135*). Excluding antibiotic-derived chemotherapies (e.g., doxorubicin), there is circumstantial and conflicting evidence for the use of antibiotics in solid tumors. Several studies in lung, colon, and pancreatic cancer suggest that eliminating intratumoral microbiota can check tumor-promoting inflammatory processes, reduce

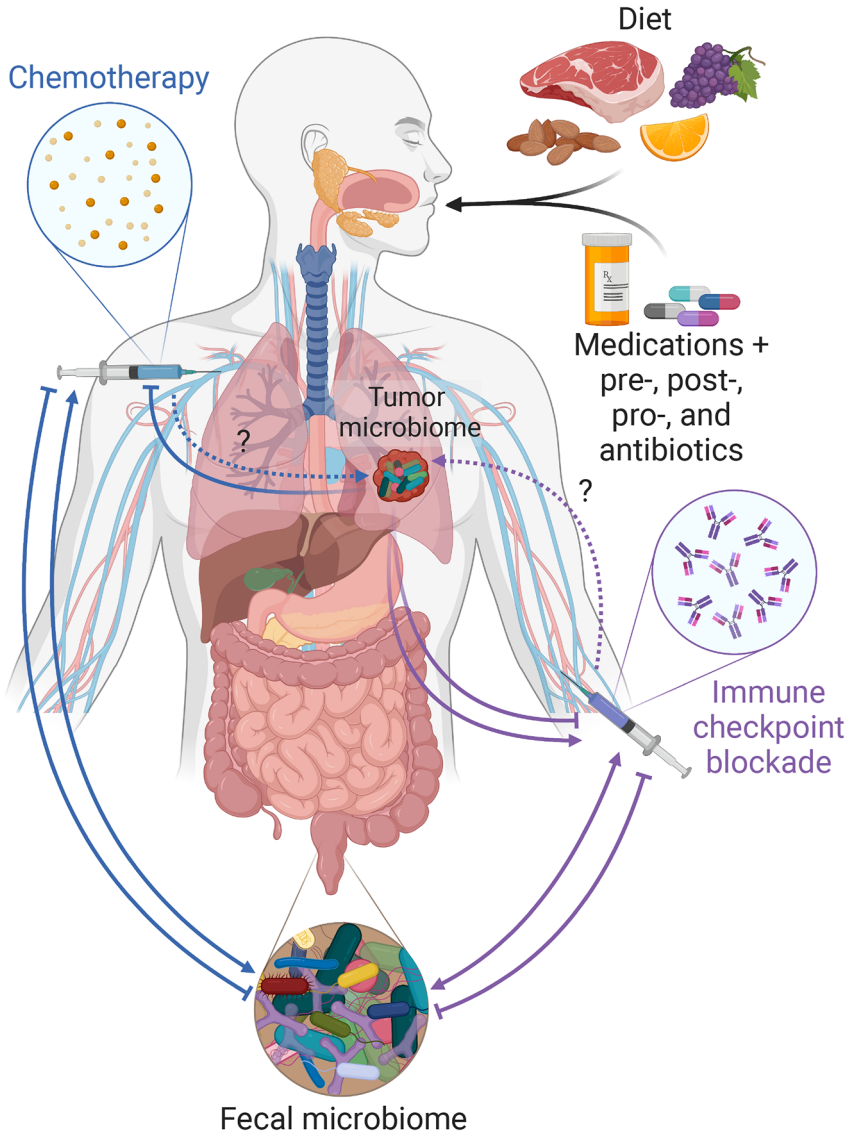


Fig. 4. Considerations when modulating the endogenous cancer microbiome. Diet, medications, and prebiotics, postbiotics, probiotics, and antibiotics all have the capacity to modify the gut and tumor microbiomes. Bidirectional influences may exist between these microbiomes and cancer therapies (chemotherapy and immunotherapy). For instance, chemotherapy can cause compositional changes in the gut microbiome, which in turn enhance treatment efficacy (*75*); in other cases, chemotherapy may be degraded by microbes (*14*). Thus, modification of the gut and/or tumor microbiomes may be advantageous for one modality of therapy while disadvantageous for another. Dotted arrows denote gaps in the literature.

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cellular proliferation, or convert a tolerogenic TME to an immunogenic one (*18*, *19*, *23*, *46*). However, increasing clinical evidence suggests that systemic antibiotics abolish immune checkpoint blockade efficacy and decrease patient survival (*133*, *136*, *137*). In hematologic malignancies, preclinical evidence suggests a careful balance, where either antibiotics or gut bacterial translocation can trigger leukemic progression in genetically predisposed hosts (*47*, *92*).

Prebiotics, postbiotics, and dietary interventions to modify the microbiome are also promising. Dietary effects on cancer were recently reviewed in detail, with many epidemiological associations but few causal mechanisms (*138*). Difficulties in dietary data collection have impeded strong conclusions, but metabolomic data that can reveal dietary intake and concomitant small-molecule effectors may help in the future. Prebiotics (molecules that promote growth of beneficial microbes) such as resistant starch, inulin, and mucin are promising in preclinical models, improving antitumor immunity and therapy response in melanoma and colon cancer (*87*), and are in clinical trials (e.g., NCT03870607, NCT03950635). Experimental evidence of postbiotic compounds (microbial-derived molecules) is limited in cancer, but they may provide advantages through defined composition and manufacturing reproducibility (*139*).

Gut microbiota can also be modulated in cancer through fecal microbiota transplantation (FMT), administration of defined microbial consortia, and commercial probiotics. FMT treats *Clostridium difficile* (now *Clostridioides difficile*) colitis effectively (*140*), with some efficacy in the treatment of immunotherapy-associated colitis (*141*). The long-term efficacy and stability of FMT remain unknown (*142*). Targeting gut microbes clinically is complicated by factors such as antibiotic preconditioning, administration route, frequency of modulation, and dietary recommendations (*142*). Ongoing clinical trials suggest that FMT from donors responsive to immunotherapy may enhance antitumor immune and potentially clinical responses [NCT03353402 (*143*)]. Additional clinical trials are evaluating the impact of transferring microbial consortia, ranging in complexity from monoclonal bacterial strains to multiplexed consortia. Few commercially available probiotic formulations have been tested for impacts on antitumor and systemic immunity, with certain formulations actually increasing tumorigenesis (*144*). In critically ill patients, commercial probiotic use may even cause bacteremia (*145*). Therefore, indiscriminate administration of commercially available probiotics in cancer patients should be discouraged.

Cancer therapy using exogenous microbiota

Major strides have been made toward engineering exogenous bacterial and viral agents for cancer therapy, particularly as powerful immunotherapy options or neoadjuvants. Two such agents have U.S. Food and Drug Administration approval: oncolytic viral therapy for advanced melanoma using talimogene laherparepvec (T-VEC) (*146*) and bacterial cancer therapy for high-risk, nonmuscle invasive bladder cancer using live-attenuated *Mycobacterium bovis* (BCG vaccine) (*147*). Because oncolytic viruses are noncommensals and have been reviewed elsewhere in detail (*148*–*150*), we focus our attention on bacterial cancer therapies (BCTs). Though historically contentious, BCT is regaining attention through synthetic biology techniques that programmatically limit systemic toxicities while enhancing regional antitumor immunity (*105*, *106*). Regulatory challenges for BCT agents are considerable (Fig. 5A), and despite ongoing clinical trials (e.g., NCT04167137), they have yet to be commercially surmounted (*151*).

Engineered microbes as cancer drugs

Natural bacterial mechanisms for tumor tropism are numerous (Fig. 5B), with intratumoral or intravenous injection often leading to ~10,000-fold accumulation in tumors relative to matched liver, spleen, and lung tissues

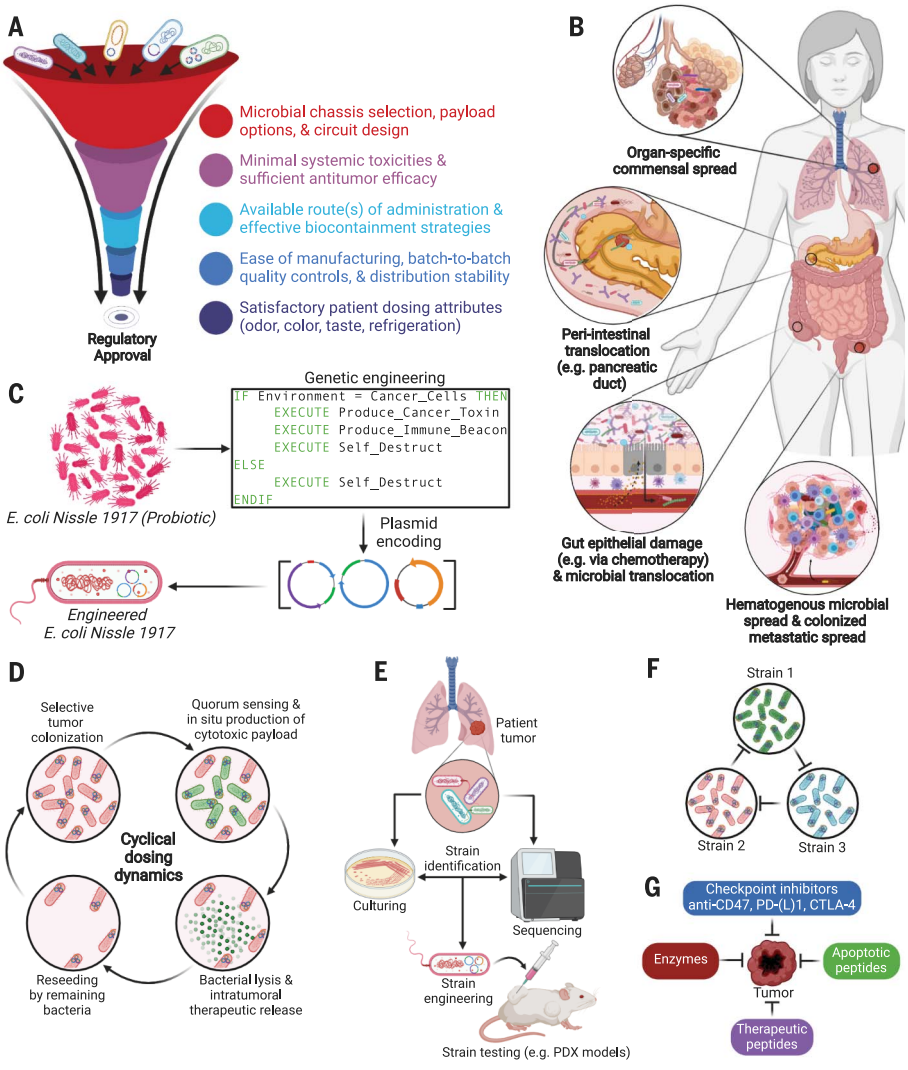


Fig. 5. Synthetic biology for exogenous cancer therapeutics. (**A**) Regulatory considerations for engineering bacteria against cancer (*151*). (**B**) Diverse sources of intratumoral bacteria include organ-specific commensals (*18*, *19*, *25*, *46*, *112*), gut communication (*15*, *20*, *23*), hematogenous spread (*172*, *173*), and intrametastatic spread (*19*). (**C**) Some probiotics, such as *E. coli* Nissle 1917, possess strong safety records (*174*), have been shown to naturally migrate to solid tumors in animal models, and can be programmed to produce and deliver therapies from within solid tumors (*155*). (**D**) Complex population dynamics can be engineered to generate the cyclical delivery of therapeutics (*155*, *156*, *175*). (**E**) Future efforts will likely center on engineering and testing strains that are found naturally in patient-specific tumors. PDX, patient-derived xenograft. (**F**) Engineered ecologies can be designed to create tailored, tumor-specific therapeutic cocktails (*176*, *177*). (**G**) Multiple drug payloads can be encoded by one or more engineered strains against tumors.

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(*152*, *153*). This affinity for tumor tissue provides a creative drug chassis and natural bridge from synthetic biology to cancer therapies, whereby cytotoxic payloads can be encoded for programmed delivery by tumor-homing bacteria (Figs. 5, C to G). Thus far, genetically attenuated, auxotrophic, and inducible versions of *Escherichia*, *Bifidobacterium*, *Listeria*, *Shigella*, *Clostridium*, *Lactococcus*, *Vibrio*, and *Salmonella* species have been engineered and have shown antitumor efficacy in preclinical models with intravenous, intratumor, and oral delivery routes (*147*). Although some approaches are based on intracellular delivery of drugs through phagocytic uptake of bacteria, others program bacteria to act as "intratumoral bioreactors" that continuously produce and release payloads extracellularly as part of colonization. An interesting general approach implements engineered bacterial lysis, which enables antitumor protein production or release only when a predefined population density of bacteria is reached (*154*–*157*). This dramatically reduces bacterial colony size and prevents systemic toxicities. Din and colleagues were the first to demonstrate how nonpathogenic *E. coli* and *Salmonella* could be engineered to lyse at a threshold population density, releasing a chemokine, hemolysin, or pro-apoptotic protein, or all three, into the TME at desired periodic intervals (*155*). The drugs are delivered cyclically as the bacterial population is programmed to generate growth-death-regrowth cycles. Chowdhury *et al*. then used this design to produce and release an antibody-fragment nanobody against CD47, which tumors can overexpress to inhibit DC phagocytosis (*156*). Intriguingly, this stimulated a tumor antigen–specific $CD8^+$ T cell response that prevented metastasis and mediated an abscopal effect, which regressed distal noninjected tumors as well. This approach further precluded host anemias and thrombocytopenias that are usually seen with systemic CD47 antagonism, suggesting a clinical opportunity. If intratumoral bacteria prove to be prevalent across various cancer types, lysis circuit designs may also provide an opportunity to flexibly engineer patient-specific, tumor-specific commensal strains (Fig. 5E) or several strains in feedback with each other (Fig. 5F) to regulate payload release. Given the many encodable cytotoxic payloads (Figs. 5G), a clear demonstration of BCT clinical efficacy with minimal systemic toxicities could considerably increase the cancer therapy armamentarium.

Outlook for the cancer microbiome

The past 15 years of microbiome research provide intriguing, though still controversial, evidence of the relationships between microbes and cancer and the nuances of these relationships. Few microbes directly cause cancer, but many more seem complicit, and, perhaps counterintuitively, several promote host antitumor immunity. This complexity may reflect shared evolutionary dynamics between the host's immune system, its commensal microbiota, and tumorigenic processes that we are just beginning to uncover (*158*–*160*).

A substantial literature gap still separates clinical observations and clinical interventions targeted at microbiota in cancer. Although gut microbiota modulation in murine immunotherapy models provides tantalizing results, they have not yet translated to commercial therapeutic interventions in humans. Moreover, observations in humans and mice of gut microbes that stratify therapy response, particularly immunotherapy (*16*, *17*, *21*, *26*), have not uncommonly shown varying taxonomic differences that persist despite uniform bioinformatic reanalyses, although there is greater concordance when examining functional profiles (*161*, *162*). Thus, many of the key problems that plagued researchers in the early 20th century—contamination, irreproducibility, patient toxicities—remain challenges today for microbially based cancer diagnostics, prognostics, and exogenous microbial therapeutics. Additional cohorts with carefully curated samples to limit and mitigate potential contamination are needed to help characterize and understand the impact of intratumoral microbes on carcinogenesis, cancer progression, and therapy response. Other efforts are needed to examine nonbacterial relationships with cancer, gastrointestinally and intratumorally, and their functions, particularly in relationship with known bacterial functions. Further consortium-level efforts are necessary to assess the quantitative impact of technical variables (e.g., DNA extraction, sample handling, bioinformatic protocols) on cancer microbiome data and guide the selection of "gold-standard" pipelines, analogous to the Microbiome Quality Control consortium's analysis of fecal amplicon sequencing among 15 laboratories and nine bioinformatic protocols (*163*).

Many of these challenges would be aided by a multicenter, longitudinal, concerted effort to study microbiota in cancer, analogous to TCGA's role in elucidating the somatic mutation landscape, with joint tumor, blood, and stool collection; multi-omic data generation; and incorporation of experimental contamination controls (Fig. 6) (*164*). Concurrent meta-analyses of existent cancer datasets with uniform in silico host depletion, decontamination, taxonomy calling, and functional profiling may be able to identify global microbial drivers in cancer pathogenesis and treatment despite technical variation between individual studies (*118*, *119*, *165*–*167*). Completion of microbiota modulation trials are additionally crucial for guiding clinical applications and increasing the cancer therapy armamentarium (*142*), with new evidence demonstrating that modulation of the gut microbiota using FMT in immunotherapy-refractory melanoma patients is associated with clinical responses and changes in the gut and tumor immune microenvironment (*143*, *168*). In-depth functional analyses at community and per-microbe scales are likely necessary to elucidate microbial-immune-cancer cell mechanistic interactions, and emerging spatial multi-omic tools may prove invaluable here (*169*, *170*). Engineered organoids with immune and microbiota niches or metabolites may further help validate or refute microbial causality or complicity in carcinogenesis, as recently demonstrated by colibactin mutagenesis studies (*22*, *171*). Although many challenges remain, building a

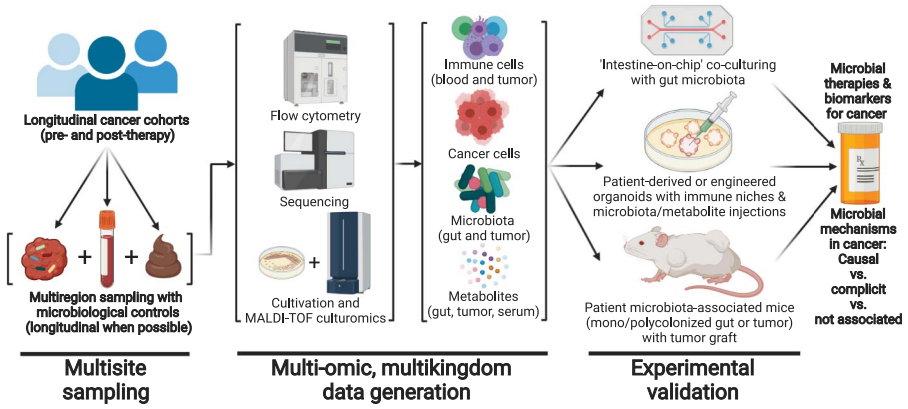


Fig. 6. Study design for characterizing cancer-associated microbiota and their functional impacts. Opportunities exist to perform large-scale identification of the presence and function of cancer-associated microbiota, beginning with longitudinal cohorts and multiregion sampling. Existing tools can be used to gather multi-omic information on host immune cells, cancer cells, microbiota, and metabolites (*51*, *178*, *179*). In vitro and in vivo disease models of a patient's tumor and intestine can then be used to verify or rebut the predicted functional impact and mechanism(s) of a given microbe (or its metabolites) and its causality in carcinogenesis (*160*, *171*). MALDI-TOF, matrix-assisted laser desorption ionization–time-of-flight mass spectrometry.

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better understanding of the roles of microbes in cancer may enable a powerful new toolkit for improving patient care.

REFERENCES AND NOTES

1. B. Ebbell, *The Papyrus Ebers: The Greatest Egyptian Medical Document* (Levin & Munksgaard, 1937).
2. S. A. Hoption Cann, J. P. van Netten, C. van Netten, Dr William Coley and tumour regression: A place in history or in the future. *Postgrad. Med. J.* **79**, 672–680 (2003). pmid: 14707241
3. W. Busch, Aus der Sitzung der medicinischen Section vom 13 November 1867. *Berl Klin Wochenschr* **5**, 137 (1868).
4. F. Fehleisen, Ueber die Züchtung der Erysipelkokken auf künstlichem Nährboden und ihre Übertragbarkeit auf den Menschen. *Dtsch. Med. Wochenschr.* **8**, 553–554 (1882). doi: 10.1055/s-0029-1196806
5. C. O. Starnes, Coley's toxins in perspective. *Nature* **357**, 11–12 (1992). doi: 10.1038/357011a0; pmid: 1574121
6. T. J. Glover, The bacteriology of cancer. *Canada Lancet Pract.* **75**, 92–111 (1930).
7. Livingston-Wheeler therapy. *CA Cancer J. Clin.* **40**, 103–108 (1990). doi: 10.3322/canjclin.40.2.103; pmid: 2106368
8. P. Rous, A sarcoma of the fowl transmissible by an agent separable from the tumor cells. *J. Exp. Med.* **13**, 397–411 (1911). doi: 10.1084/jem.13.4.397; pmid: 19867421
9. M. K. White, J. S. Pagano, K. Khalili, Viruses and human cancers: A long road of discovery of molecular paradigms. *Clin. Microbiol. Rev.* **27**, 463–481 (2014). doi: 10.1128/CMR.00124-13; pmid: 24982317
10. D. Hanahan, R. A. Weinberg, The hallmarks of cancer. *Cell* **100**, 57–70 (2000). doi: 10.1016/S0092-8674(00)81683-9; pmid: 10647931
11. D. Hanahan, R. A. Weinberg, Hallmarks of cancer: The next generation. *Cell* **144**, 646–674 (2011). doi: 10.1016/j.cell.2011.02.013; pmid: 21376230
12. D. Nejman *et al*., The human tumor microbiome is composed of tumor type-specific intracellular bacteria. *Science* **368**, 973–980 (2020). doi: 10.1126/science.aay9189; pmid: 32467386
13. G. D. Poore *et al*., RETRACTED ARTICLE: Microbiome analyses of blood and tissues suggest cancer diagnostic approach. *Nature* **579**, 567–574 (2020). doi: 10.1038/s41586-020-2095-1; pmid: 32214244
14. L. T. Geller *et al*., Potential role of intratumor bacteria in mediating tumor resistance to the chemotherapeutic drug gemcitabine. *Science* **357**, 1156–1160 (2017). doi: 10.1126/science.aah5043; pmid: 28912244
15. E. Riquelme *et al*., Tumor microbiome diversity and composition influence pancreatic cancer outcomes. *Cell* **178**, 795–806.e12 (2019). doi: 10.1016/j.cell.2019.07.008; pmid: 31398337
16. V. Gopalakrishnan *et al*., Gut microbiome modulates response to anti–PD-1 immunotherapy in melanoma patients. *Science* **359**, 97–103 (2018). doi: 10.1126/science.aan4236; pmid: 29097493
17. B. Routy *et al*., Gut microbiome influences efficacy of PD-1–based immunotherapy against epithelial tumors. *Science* **359**, 91–97 (2018). doi: 10.1126/science.aan3706; pmid: 29097494
18. C. Jin *et al*., Commensal microbiota promote lung cancer development via γδ T cells. *Cell* **176**, 998–1013.e16 (2019). doi: 10.1016/j.cell.2018.12.040; pmid: 30712876
19. S. Bullman *et al*., Analysis of *Fusobacterium* persistence and antibiotic response in colorectal cancer. *Science* **358**, 1443–1448 (2017). doi: 10.1126/science.aal5240; pmid: 29170280
20. B. Aykut *et al*., The fungal mycobiome promotes pancreatic oncogenesis via activation of MBL. *Nature* **574**, 264–267 (2019). doi: 10.1038/s41586-019-1608-2; pmid: 31578522
21. V. Matson *et al*., The commensal microbiome is associated with anti–PD-1 efficacy in metastatic melanoma patients. *Science* **359**, 104–108 (2018). doi: 10.1126/science.aao3290; pmid: 29302014
22. C. Pleguezuelos-Manzano *et al*., Mutational signature in colorectal cancer caused by genotoxic *pks*$^{+}$ *E. coli*. *Nature* **580**, 269–273 (2020). doi: 10.1038/s41586-020-2080-8; pmid: 32106218
23. S. Pushalkar *et al*., The pancreatic cancer microbiome promotes oncogenesis by induction of innate and adaptive immune suppression. *Cancer Discov.* **8**, 403–416 (2018). doi: 10.1158/2159-8290.CD-17-1134; pmid: 29567829
24. L. E. Fulbright, M. Ellermann, J. C. Arthur, The microbiome and the hallmarks of cancer. *PLOS Pathog.* **13**, e1006480 (2017). doi: 10.1371/journal.ppat.1006480; pmid: 28934351
25. J. J. Tsay *et al*., Lower airway dysbiosis affects lung cancer progression. *Cancer Discov.* **11**, 293–307 (2021). doi: 10.1158/2159-8290.CD-20-0263; pmid: 33177060
26. A. E. Frankel *et al*., Metagenomic shotgun sequencing and unbiased metabolomic profiling identify specific human gut microbiota and metabolites associated with immune checkpoint therapy efficacy in melanoma patients. *Neoplasia* **19**, 848–855 (2017). doi: 10.1016/j.neo.2017.08.004; pmid: 28923537
27. R. Sender, S. Fuchs, R. Milo, Revised estimates for the number of human and bacteria cells in the body. *PLOS Biol.* **14**, e1002533 (2016). doi: 10.1371/journal.pbio.1002533; pmid: 27541692
28. Y. Belkaid, S. Naik, Compartmentalized and systemic control of tissue immunity by commensals. *Nat. Immunol.* **14**, 646–653 (2013). doi: 10.1038/ni.2604; pmid: 23778791
29. T. C. Fung, C. A. Olson, E. Y. Hsiao, Interactions between the microbiota, immune and nervous systems in health and disease. *Nat. Neurosci.* **20**, 145–155 (2017). doi: 10.1038/nn.4476; pmid: 28092661
30. L. Zitvogel, Y. Ma, D. Raoult, G. Kroemer, T. F. Gajewski, The microbiome in cancer immunotherapy: Diagnostic tools and therapeutic strategies. *Science* **359**, 1366–1370 (2018). doi: 10.1126/science.aar6918; pmid: 29567708
31. W. S. Garrett, Cancer and the microbiota. *Science* **348**, 80–86 (2015). doi: 10.1126/science.aaa4972; pmid: 25838377
32. V. Gopalakrishnan, B. A. Helmink, C. N. Spencer, A. Reuben, J. A. Wargo, The influence of the gut microbiome on cancer, immunity, and cancer immunotherapy. *Cancer Cell* **33**, 570–580 (2018). doi: 10.1016/j.ccell.2018.03.015; pmid: 29634945
33. B. A. Helmink, M. A. W. Khan, A. Hermann, V. Gopalakrishnan, J. A. Wargo, The microbiome, cancer, and cancer therapy. *Nat. Med.* **25**, 377–388 (2019). doi: 10.1038/s41591-019-0377-7; pmid: 30842679
34. L. Parhi *et al*., Breast cancer colonization by *Fusobacterium nucleatum* accelerates tumor growth and metastatic progression. *Nat. Commun.* **11**, 3259 (2020). doi: 10.1038/s41467-020-16967-2; pmid: 32591509
35. K. J. Locey, J. T. Lennon, Scaling laws predict global microbial diversity. *Proc. Natl. Acad. Sci. U.S.A.* **113**, 5970–5975 (2016). doi: 10.1073/pnas.1521291113; pmid: 27140646
36. IARC Working Group on the Evaluation of Carcinogenic Risks to Humans, Biological agents. Volume 100 B. A review of human carcinogens. *IARC Monogr. Eval. Carcinog. Risks Hum.* **100**, 1–441 (2012). pmid: 23189750
37. M. R. Wilson *et al*., The human gut bacterial genotoxin colibactin alkylates DNA. *Science* **363**, eaar7785 (2019). doi: 10.1126/science.aar7785; pmid: 30765538
38. C. M. Dejea *et al*., Patients with familial adenomatous polyposis harbor colonic biofilms containing tumorigenic bacteria. *Science* **359**, 592–597 (2018). doi: 10.1126/science.aah3648; pmid: 29420293
39. Z. He *et al*., *Campylobacter jejuni* promotes colorectal tumorigenesis through the action of cytolethal distending toxin. *Gut* **68**, 289–300 (2019). doi: 10.1136/gutjnl-2018-317200; pmid: 30377189
40. A. Boot *et al*., Characterization of colibactin-associated mutational signature in an Asian oral squamous cell carcinoma and in other mucosal tumor types. *Genome Res.* **30**, 803–813 (2020). doi: 10.1101/gr.255620.119; pmid: 32661091
41. M. Barrett, C. K. Hand, F. Shanahan, T. Murphy, P. W. O'Toole, Mutagenesis by microbe: The role of the microbiota in shaping the cancer genome. *Trends Cancer* **6**, 277–287 (2020). doi: 10.1016/j.trecan.2020.01.019; pmid: 32209443
42. O. Silva-García, J. J. Valdez-Alarcón, V. M. Baizabal-Aguirre, Wnt/β-catenin signaling as a molecular target by pathogenic bacteria. *Front. Immunol.* **10**, 2135 (2019). doi: 10.3389/fimmu.2019.02135; pmid: 31611869
43. M. R. Rubinstein *et al*., *Fusobacterium nucleatum* promotes colorectal carcinogenesis by modulating E-cadherin/β-catenin signaling via its FadA adhesin. *Cell Host Microbe* **14**, 195–206 (2013). doi: 10.1016/j.chom.2013.07.012; pmid: 23954158
44. E. Kadosh *et al*., The gut microbiome switches mutant p53 from tumour-suppressive to oncogenic. *Nature* **586**, 133–138 (2020). doi: 10.1038/s41586-020-2541-0; pmid: 32728212
45. C. Ma *et al*., Gut microbiome-mediated bile acid metabolism regulates liver cancer via NKT cells. *Science* **360**, eaan5931 (2018). doi: 10.1126/science.aan5931; pmid: 29798856
46. V. Le Noci *et al*., Modulation of pulmonary microbiota by antibiotic or probiotic aerosol therapy: A strategy to promote immunosurveillance against lung metastases. *Cell Rep.* **24**, 3528–3538 (2018). doi: 10.1016/j.celrep.2018.08.090; pmid: 30257213
47. M. Meisel *et al*., Microbial signals drive pre-leukaemic myeloproliferation in a Tet2-deficient host. *Nature* **557**, 580–584 (2018). doi: 10.1038/s41586-018-0125-z; pmid: 29769727
48. A. Almeida *et al*., A new genomic blueprint of the human gut microbiota. *Nature* **568**, 499–504 (2019). doi: 10.1038/s41586-019-0965-1; pmid: 30745586
49. M. B. Dion, F. Oechslin, S. Moineau, Phage diversity, genomics and phylogeny. *Nat. Rev. Microbiol.* **18**, 125–138 (2020). doi: 10.1038/s41579-019-0311-5; pmid: 32015529
50. B. Chassaing, M. Kumar, M. T. Baker, V. Singh, M. Vijay-Kumar, Mammalian gut immunity. *Biomed. J.* **37**, 246–258 (2014). doi: 10.4103/2319-4170.130922; pmid: 25163502
51. ICGC/TCGA Pan-ancer Analysis of Whole Genomes Consortium, Pan-cancer analysis of whole genomes. *Nature* **578**, 82–93 (2020). doi: 10.1038/s41586-020-1969-6; pmid: 32025007
52. U. Del Monte, Does the cell number 10^{9} still really fit one gram of tumor tissue? *Cell Cycle* **8**, 505–506 (2009). doi: 10.4161/cc.8.3.7608; pmid: 19176997
53. G. Erdag *et al*., Immunotype and immunohistologic characteristics of tumor-infiltrating immune cells are associated with clinical outcome in metastatic melanoma. *Cancer Res.* **72**, 1070–1080 (2012). doi: 10.1158/0008-5472.CAN-11-3218; pmid: 22266112
54. J. N. Kather *et al*., Topography of cancer-associated immune cells in human solid tumors. *eLife* **7**, e36967 (2018). doi: 10.7554/eLife.36967; pmid: 30179157
55. L. Zitvogel, M. Ayyoub, B. Routy, G. Kroemer, Microbiome and anticancer immunosurveillance. *Cell* **165**, 276–287 (2016). doi: 10.1016/j.cell.2016.03.001; pmid: 27058662
56. F. V. Michelis *et al*., Early lymphocyte recovery at 28 d post-transplant is predictive of reduced risk of relapse in patients with acute myeloid leukemia transplanted with peripheral blood stem cell grafts. *Eur. J. Haematol.* **93**, 273–280 (2014). doi: 10.1111/ejh.12338; pmid: 24725056
57. A. Staffas *et al*., Nutritional support from the intestinal microbiota improves hematopoietic reconstitution after bone marrow transplantation in mice. *Cell Host Microbe* **23**, 447–457.e4 (2018). doi: 10.1016/j.chom.2018.03.002; pmid: 29576480
58. J. U. Peled *et al*., Microbiota as predictor of mortality in allogeneic hematopoietic-cell transplantation. *N. Engl. J. Med.* **382**, 822–834 (2020). doi: 10.1056/NEJMoa1900623; pmid: 32101664
59. J. Schluter *et al*., The gut microbiota is associated with immune cell dynamics in humans. *Nature* **588**, 303–307 (2020). doi: 10.1038/s41586-020-2971-8; pmid: 33239790
60. I. Brook, G. D. Ledney, Effect of antimicrobial therapy on the gastrointestinal bacterial flora, infection and mortality in mice exposed to different doses of irradiation. *J. Antimicrob. Chemother.* **33**, 63–72 (1994). doi: 10.1093/jac/33.1.63; pmid: 8157575
61. L. G. Burdelya *et al*., An agonist of toll-like receptor 5 has radioprotective activity in mouse and primate models. *Science* **320**, 226–230 (2008). doi: 10.1126/science.1154986; pmid: 18403709
62. W. Liu *et al*., Radioprotector WR-2721 and mitigating peptidoglycan synergistically promote mouse survival through the amelioration of intestinal and bone marrow damage. *J. Radiat. Res. (Tokyo)* **56**, 278–286 (2015). doi: 10.1093/jrr/rru100; pmid: 25617317
63. H. Guo *et al*., Multi-omics analyses of radiation survivors identify radioprotective microbes and metabolites. *Science* **370**, eaay9097 (2020). doi: 10.1126/science.aay9097; pmid: 33122357
64. A. Khosravi *et al*., Gut microbiota promote hematopoiesis to control bacterial infection. *Cell Host Microbe* **15**, 374–381 (2014). doi: 10.1016/j.chom.2014.02.006; pmid: 24629343
65. T. B. Clarke *et al*., Recognition of peptidoglycan from the microbiota by Nod1 enhances systemic innate immunity. *Nat. Med.* **16**, 228–231 (2010). doi: 10.1038/nm.2087; pmid: 20081863
66. J. C. Fischer *et al*., RIG-I/MAVS and STING signaling promote gut integrity during irradiation- and immune-mediated tissue injury. *Sci. Transl. Med.* **9**, eaag2513 (2017). doi: 10.1126/scitranslmed.aag2513; pmid: 28424327

Downloaded from https://www.science.org at University of California San Diego on March 11, 2026

ERRATUM PUBLISHED 26 SEPTEMBER 2024; SEE LAST PAGE

67. K. Honda, D. R. Littman, The microbiota in adaptive immune homeostasis and disease. *Nature* **535**, 75–84 (2016). doi: 10.1038/nature18848; pmid: 27383982
68. S. Viaud *et al*., The intestinal microbiota modulates the anticancer immune effects of cyclophosphamide. *Science* **342**, 971–976 (2013). doi: 10.1126/science.1240537; pmid: 24264990
69. R. Daillère *et al*., *Enterococcus hirae* and *Barnesiella intestinihominis* facilitate cyclophosphamide-induced therapeutic immunomodulatory effects. *Immunity* **45**, 931–943 (2016). doi: 10.1016/j.immuni.2016.09.009; pmid: 27717798
70. M. Vétizou *et al*., Anticancer immunotherapy by CTLA-4 blockade relies on the gut microbiota. *Science* **350**, 1079–1084 (2015). doi: 10.1126/science.aad1329; pmid: 26541610
71. A. Sivan *et al*., Commensal *Bifidobacterium* promotes antitumor immunity and facilitates anti–PD-L1 efficacy. *Science* **350**, 1084–1089 (2015). doi: 10.1126/science.aac4255; pmid: 26541606
72. M. Uribe-Herranz *et al*., Gut microbiota modulate dendritic cell antigen presentation and radiotherapy-induced antitumor immune response. *J. Clin. Invest.* **130**, 466–479 (2020). doi: 10.1172/JCI124332; pmid: 31815742
73. C. A. Bessell *et al*., Commensal bacteria stimulate antitumor responses via T cell cross-reactivity. *JCI Insight* **5**, e135597 (2020). doi: 10.1172/jci.insight.135597; pmid: 32324171
74. C. M. Paulos *et al*., Microbial translocation augments the function of adoptively transferred self/tumor-specific $CD8^+$ T cells via TLR4 signaling. *J. Clin. Invest.* **117**, 2197–2204 (2007). doi: 10.1172/JCI32205; pmid: 17657310
75. M. P. Roberti *et al*., Chemotherapy-induced ileal crypt apoptosis and the ileal microbiome shape immunosurveillance and prognosis of proximal colon cancer. *Nat. Med.* **26**, 919–931 (2020). doi: 10.1038/s41591-020-0882-8; pmid: 32451498
76. Y. Shi *et al*., Intratumoral accumulation of gut microbiota facilitates CD47-based immunotherapy via STING signaling. *J. Exp. Med.* **217**, e20192282 (2020). doi: 10.1084/jem.20192282; pmid: 32142585
77. Y. Li *et al*., Gut microbiota dependent anti-tumor immunity restricts melanoma growth in $Rnf5^{-/-}$ mice. *Nat. Commun.* **10**, 1492 (2019). doi: 10.1038/s41467-019-09525-y; pmid: 30940817
78. E. Ansaldo *et al*., *Akkermansia muciniphila* induces intestinal adaptive immune responses during homeostasis. *Science* **364**, 1179–1184 (2019). doi: 10.1126/science.aaw7479; pmid: 31221858
79. C. Gil-Cruz *et al*., Microbiota-derived peptide mimics drive lethal inflammatory cardiomyopathy. *Science* **366**, 881–886 (2019). doi: 10.1126/science.aav3487; pmid: 31727837
80. R. Hebbandi Nanjundappa *et al*., A gut microbial mimic that hijacks diabetogenic autoreactivity to suppress colitis. *Cell* **171**, 655–667.e17 (2017). doi: 10.1016/j.cell.2017.09.022; pmid: 29053971
81. J. Petersen *et al*., T cell receptor cross-reactivity between gliadin and bacterial peptides in celiac disease. *Nat. Struct. Mol. Biol.* **27**, 49–61 (2020). doi: 10.1038/s41594-019-0353-4; pmid: 31873306
82. A. Snyder *et al*., Genetic basis for clinical response to CTLA-4 blockade in melanoma. *N. Engl. J. Med.* **371**, 2189–2199 (2014). doi: 10.1056/NEJMoa1406498; pmid: 25409260
83. A. Fluckiger *et al*., Cross-reactivity between tumor MHC class I–restricted antigens and an enterococcal bacteriophage. *Science* **369**, 936–942 (2020). doi: 10.1126/science.aax0701; pmid: 32820119
84. T. Tanoue *et al*., A defined commensal consortium elicits CD8 T cells and anti-cancer immunity. *Nature* **565**, 600–605 (2019). doi: 10.1038/s41586-019-0878-z; pmid: 30675064
85. C. Coutzac *et al*., Systemic short chain fatty acids limit antitumor effect of CTLA-4 blockade in hosts with cancer. *Nat. Commun.* **11**, 2168 (2020). doi: 10.1038/s41467-020-16079-x; pmid: 32358520
86. M. Nomura *et al*., Association of short-chain fatty acids in the gut microbiome with clinical response to treatment with nivolumab or pembrolizumab in patients with solid cancer tumors. *JAMA Netw. Open* **3**, e202895 (2020). doi: 10.1001/jamanetworkopen.2020.2895; pmid: 32297948
87. Y. Li *et al*., Prebiotic-induced anti-tumor immunity attenuates tumor growth. *Cell Rep.* **30**, 1753–1766.e6 (2020). doi: 10.1016/j.celrep.2020.01.035; pmid: 32049008
88. O. Y. Kim *et al*., Bacterial outer membrane vesicles suppress tumor by interferon-γ-mediated antitumor response. *Nat. Commun.* **8**, 626 (2017). doi: 10.1038/s41467-017-00729-8; pmid: 28931823
89. N. Singh *et al*., Activation of Gpr109a, receptor for niacin and the commensal metabolite butyrate, suppresses colonic inflammation and carcinogenesis. *Immunity* **40**, 128–139 (2014). doi: 10.1016/j.immuni.2013.12.007; pmid: 24412617
90. N. Iida *et al*., Commensal bacteria control cancer response to therapy by modulating the tumor microenvironment. *Science* **342**, 967–970 (2013). doi: 10.1126/science.1240527; pmid: 24264989
91. L. Wang *et al*., A purified membrane protein from *Akkermansia muciniphila* or the pasteurised bacterium blunts colitis associated tumourigenesis by modulation of $CD8^+$ T cells in mice. *Gut* **69**, 1988–1997 (2020). doi: 10.1136/gutjnl-2019-320105; pmid: 32169907
92. C. Vicente-Dueñas *et al*., An intact gut microbiome protects genetically predisposed mice against leukemia. *Blood* **136**, 2003–2017 (2020). doi: 10.1182/blood.2019004381; pmid: 32911536
93. L. F. Mager *et al*., Microbiome-derived inosine modulates response to checkpoint inhibitor immunotherapy. *Science* **369**, 1481–1489 (2020). doi: 10.1126/science.abc3421; pmid: 32792462
94. M. Amit *et al*., Loss of p53 drives neuron reprogramming in head and neck cancer. *Nature* **578**, 449–454 (2020). doi: 10.1038/s41586-020-1996-3; pmid: 32051587
95. V. Venkataramani *et al*., Glutamatergic synaptic input to glioma cells drives brain tumour progression. *Nature* **573**, 532–538 (2019). doi: 10.1038/s41586-019-1564-x; pmid: 31534219
96. Q. Zeng *et al*., Synaptic proximity enables NMDAR signalling to promote brain metastasis. *Nature* **573**, 526–531 (2019). doi: 10.1038/s41586-019-1576-6; pmid: 31534217
97. H. S. Venkatesh *et al*., Electrical and synaptic integration of glioma into neural circuits. *Nature* **573**, 539–545 (2019). doi: 10.1038/s41586-019-1563-y; pmid: 31534222
98. P. A. Muller *et al*., Microbiota-modulated $CART^+$ enteric neurons autonomously regulate blood glucose. *Science* **370**, 314–321 (2020). doi: 10.1126/science.abd6176; pmid: 32855216
99. P. Spanogiannopoulos, E. N. Bess, R. N. Carmody, P. J. Turnbaugh, The microbial pharmacists within us: A metagenomic view of xenobiotic metabolism. *Nat. Rev. Microbiol.* **14**, 273–287 (2016). doi: 10.1038/nrmicro.2016.17; pmid: 26972811
100. B. A. Daisley *et al*., Abiraterone acetate preferentially enriches for the gut commensal *Akkermansia muciniphila* in castrate-resistant prostate cancer patients. *Nat. Commun.* **11**, 4822 (2020). doi: 10.1038/s41467-020-18649-5; pmid: 32973149
101. K. Mima *et al*., *Fusobacterium nucleatum* and T cells in colorectal carcinoma. *JAMA Oncol.* **1**, 653–661 (2015). doi: 10.1001/jamaoncol.2015.1377; pmid: 26181352
102. T. Hamada *et al*., *Fusobacterium nucleatum* in colorectal cancer relates to immune response differentially by tumor microsatellite instability status. *Cancer Immunol. Res.* **6**, 1327–1336 (2018). doi: 10.1158/2326-6066.CIR-18-0174; pmid: 30228205
103. M. Suzuki *et al*., *Helicobacter pylori* CagA phosphorylation-independent function in epithelial proliferation and inflammation. *Cell Host Microbe* **5**, 23–34 (2009). doi: 10.1016/j.chom.2008.11.010; pmid: 19154985
104. T. Yu *et al*., *Fusobacterium nucleatum* promotes chemoresistance to colorectal cancer by modulating autophagy. *Cell* **170**, 548–563.e16 (2017). doi: 10.1016/j.cell.2017.07.008; pmid: 28753429
105. N. S. Forbes, Engineering the perfect (bacterial) cancer therapy. *Nat. Rev. Cancer* **10**, 785–794 (2010). doi: 10.1038/nrc2934; pmid: 20944664
106. S. Zhou, C. Gravekamp, D. Bermudes, K. Liu, Tumour-targeting bacteria engineered to fight cancer. *Nat. Rev. Cancer* **18**, 727–743 (2018). doi: 10.1038/s41568-018-0070-z; pmid: 30405213
107. W. S. Garrett, J. I. Gordon, L. H. Glimcher, Homeostasis and inflammation in the intestine. *Cell* **140**, 859–870 (2010). doi: 10.1016/j.cell.2010.01.023; pmid: 20303876
108. D. N. O'Dwyer, R. P. Dickson, B. B. Moore, The lung microbiome, immunity, and the pathogenesis of chronic lung disease. *J. Immunol.* **196**, 4839–4847 (2016). doi: 10.4049/jimmunol.1600279; pmid: 27260767
109. A. Stacy, Y. Belkaid, Microbial guardians of skin health. *Science* **363**, 227–228 (2019). doi: 10.1126/science.aat4326; pmid: 30655428
110. D. F. Squarzanti *et al*., Non-melanoma skin cancer: News from microbiota research. *Crit. Rev. Microbiol.* **46**, 433–449 (2020). doi: 10.1080/1040841X.2020.1794792; pmid: 32692305
111. J. Liu, M. Luo, Y. Zhang, G. Cao, S. Wang, Association of high-risk human papillomavirus infection duration and cervical lesions with vaginal microbiota composition. *Ann. Transl. Med.* **8**, 1161 (2020). doi: 10.21037/atm-20-5832; pmid: 33241010
112. K. A. So *et al*., Changes of vaginal microbiota during cervical carcinogenesis in women with human papillomavirus infection. *PLOS ONE* **15**, e0238705 (2020). doi: 10.1371/journal.pone.0238705; pmid: 32941440
113. Human Microbiome Project Consortium, Structure, function and diversity of the healthy human microbiome. *Nature* **486**, 207–214 (2012). doi: 10.1038/nature11234; pmid: 22699609
114. I. M. Campbell, C. A. Shaw, P. Stankiewicz, J. R. Lupski, Somatic mosaicism: Implications for disease and transmission genetics. *Trends Genet.* **31**, 382–392 (2015). doi: 10.1016/j.tig.2015.03.013; pmid: 25910407
115. M. H. Bailey *et al*., Comprehensive characterization of cancer driver genes and mutations. *Cell* **173**, 371–385.e18 (2018). doi: 10.1016/j.cell.2018.02.060; pmid: 29625053
116. R. S. Klein *et al*., Association of *Streptococcus bovis* with carcinoma of the colon. *N. Engl. J. Med.* **297**, 800–802 (1977). doi: 10.1056/NEJM197710132971503; pmid: 408687
117. T. N. Y. Kwong *et al*., Association between bacteremia from specific microbes and subsequent diagnosis of colorectal cancer. *Gastroenterology* **155**, 383–390.e8 (2018). doi: 10.1053/j.gastro.2018.04.028; pmid: 29729257
118. J. Wirbel *et al*., Meta-analysis of fecal metagenomes reveals global microbial signatures that are specific for colorectal cancer. *Nat. Med.* **25**, 679–689 (2019). doi: 10.1038/s41591-019-0406-6; pmid: 30936547
119. A. M. Thomas *et al*., Metagenomic analysis of colorectal cancer datasets identifies cross-cohort microbial diagnostic signatures and a link with choline degradation. *Nat. Med.* **25**, 667–678 (2019). doi: 10.1038/s41591-019-0405-7; pmid: 30936548
120. S. Yachida *et al*., Metagenomic and metabolomic analyses reveal distinct stage-specific phenotypes of the gut microbiota in colorectal cancer. *Nat. Med.* **25**, 968–976 (2019). doi: 10.1038/s41591-019-0458-7; pmid: 31171880
121. B. Flemer *et al*., Tumour-associated and non-tumour-associated microbiota in colorectal cancer. *Gut* **66**, 633–643 (2017). doi: 10.1136/gutjnl-2015-309595; pmid: 26992426
122. J. J. Farrell *et al*., Variations of oral microbiota are associated with pancreatic diseases including pancreatic cancer. *Gut* **61**, 582–588 (2012). doi: 10.1136/gutjnl-2011-300784; pmid: 21994333
123. E. Half *et al*., Fecal microbiome signatures of pancreatic cancer patients. *Sci. Rep.* **9**, 16801 (2019). doi: 10.1038/s41598-019-53041-4; pmid: 31727922
124. X. Yan *et al*., Discovery and validation of potential bacterial biomarkers for lung cancer. *Am. J. Cancer Res.* **5**, 3111–3122 (2015). pmid: 26693063
125. S. H. Lee *et al*., Characterization of microbiome in bronchoalveolar lavage fluid of patients with lung cancer comparing with benign mass like lesions. *Lung Cancer* **102**, 89–95 (2016). doi: 10.1016/j.lungcan.2016.10.016; pmid: 27987594
126. K. L. Greathouse *et al*., Interaction between the microbiome and TP53 in human lung cancer. *Genome Biol.* **19**, 123 (2018). doi: 10.1186/s13059-018-1501-6; pmid: 30143034
127. C. Urbaniak *et al*., Microbiota of human breast tissue. *Appl. Environ. Microbiol.* **80**, 3007–3014 (2014). doi: 10.1128/AEM.00242-14; pmid: 24610844
128. C. Urbaniak *et al*., The microbiota of breast tissue and its association with breast cancer. *Appl. Environ. Microbiol.* **82**, 5039–5048 (2016). doi: 10.1128/AEM.01235-16; pmid: 27342554
129. A. R. Cantwell Jr., D. W. Kelso, Microbial findings in cancers of the breast and in their metastases to the skin. Implications for etiology. *J. Dermatol. Surg. Oncol.* **7**, 483–491 (1981). doi: 10.1111/j.1524-4725.1981.tb00682.x; pmid: 6166642
130. J. I. Gorelick, L. B. Senterfit, E. D. Vaughan Jr., Quantitative bacterial tissue cultures from 209 prostatectomy specimens: Findings and implications. *J. Urol.* **139**, 57–60 (1988). doi: 10.1016/S0022-5347(17)42292-0; pmid: 3336106
131. R. J. Cohen, B. A. Shannon, J. E. McNeal, T. Shannon, K. L. Garrett, Propionibacterium acnes associated with inflammation in radical prostatectomy specimens: A possible link to cancer evolution? *J. Urol.* **173**, 1969–1974 (2005). doi: 10.1097/01.ju.0000158161.15277.78; pmid: 15879794
132. M. Castellarin *et al*., *Fusobacterium nucleatum* infection is prevalent in human colorectal carcinoma. *Genome Res.* **22**, 299–306 (2012). doi: 10.1101/gr.126516.111; pmid: 22009989

Downloaded from https://www.science.org at University of California San Diego on March 11, 2026

ERRATUM PUBLISHED 26 SEPTEMBER 2024; SEE LAST PAGE

133. D. J. Pinato *et al*., Association of prior antibiotic treatment with survival and response to immune checkpoint inhibitor therapy in patients with cancer. *JAMA Oncol.* **5**, 1774–1778 (2019). doi: 10.1001/jamaoncol.2019.2785; pmid: 31513236
134. D. R. Lowy, J. T. Schiller, Preventing cancer and other diseases caused by human papillomavirus infection: 2017 Lasker-DeBakey Clinical Research Award. *JAMA* **318**, 901–902 (2017). doi: 10.1001/jama.2017.11706; pmid: 28876435
135. M. Stolte *et al*., Helicobacter and gastric MALT lymphoma. *Gut* **50**, III19–III24 (2002). doi: 10.1136/gut.50.suppl_3.iii19; pmid: 11953328
136. X.-Z. Huang *et al*., Antibiotic use and the efficacy of immune checkpoint inhibitors in cancer patients: A pooled analysis of 2740 cancer patients. *OncoImmunology* **8**, e1665973 (2019). doi: 10.1080/2162402X.2019.1665973; pmid: 31741763
137. L. Derosa *et al*., Negative association of antibiotics on clinical activity of immune checkpoint inhibitors in patients with advanced renal cell and non-small-cell lung cancer. *Ann. Oncol.* **29**, 1437–1444 (2018). doi: 10.1093/annonc/mdy103; pmid: 29617710
138. S. E. Steck, E. A. Murphy, Dietary patterns and cancer risk. *Nat. Rev. Cancer* **20**, 125–138 (2020). doi: 10.1038/s41568-019-0227-4; pmid: 31848467
139. A. H. Rad, A. Abbasi, H. S. Kafil, K. Ganbarov, Potential pharmaceutical and food applications of postbiotics: A review. *Curr. Pharm. Biotechnol.* **21**, 1576–1587 (2020). doi: 10.2174/1389201021666200516154833; pmid: 32416671
140. E. van Nood *et al*., Duodenal infusion of donor feces for recurrent *Clostridium difficile*. *N. Engl. J. Med.* **368**, 407–415 (2013). doi: 10.1056/NEJMoa1205037; pmid: 23323867
141. Y. Wang *et al*., Fecal microbiota transplantation for refractory immune checkpoint inhibitor-associated colitis. *Nat. Med.* **24**, 1804–1808 (2018). doi: 10.1038/s41591-018-0238-9; pmid: 30420754
142. J. L. McQuade, C. R. Daniel, B. A. Helmink, J. A. Wargo, Modulating the microbiome to improve therapeutic response in cancer. *Lancet Oncol.* **20**, e77–e91 (2019). doi: 10.1016/S1470-2045(18)30952-5; pmid: 30712808
143. E. N. Baruch *et al*., Fecal microbiota transplant promotes response in immunotherapy-refractory melanoma patients. *Science* **371**, 602–609 (2021). doi: 10.1126/science.abb5920; pmid: 33303685
144. J. C. Arthur *et al*., VSL#3 probiotic modifies mucosal microbial composition but does not reduce colitis-associated colorectal cancer. *Sci. Rep.* **3**, 2868 (2013). doi: 10.1038/srep02868; pmid: 24100376
145. I. Yelin *et al*., Genomic and epidemiological evidence of bacterial transmission from probiotic capsule to blood in ICU patients. *Nat. Med.* **25**, 1728–1732 (2019). doi: 10.1038/s41591-019-0626-9; pmid: 31700189
146. R. M. Conry, B. Westbrook, S. McKee, T. G. Norwood, Talimogene laherparepvec: First in class oncolytic virotherapy. *Hum. Vaccin. Immunother.* **14**, 839–846 (2018). doi: 10.1080/21645515.2017.1412896; pmid: 29420123
147. M. G. Kramer, M. Masner, F. A. Ferreira, R. M. Hoffman, Bacterial therapy of cancer: Promises, limitations, and insights for future directions. *Front. Microbiol.* **9**, 16 (2018). doi: 10.3389/fmicb.2018.00016; pmid: 29472896
148. M. Zheng, J. Huang, A. Tong, H. Yang, Oncolytic viruses for cancer therapy: barriers and recent advances. *Mol. Ther. Oncolytics* **15**, 234–247 (2019). doi: 10.1016/j.omto.2019.10.007; pmid: 31872046
149. E. Ylösmäki, V. Cerullo, Design and application of oncolytic viruses for cancer immunotherapy. *Curr. Opin. Biotechnol.* **65**, 25–36 (2020). doi: 10.1016/j.copbio.2019.11.016; pmid: 31874424
150. P. K. Bommareddy, M. Shettigar, H. L. Kaufman, Integrating oncolytic viruses in combination cancer immunotherapy. *Nat. Rev. Immunol.* **18**, 498–513 (2018). doi: 10.1038/s41577-018-0014-6; pmid: 29743717
151. M. R. Charbonneau, V. M. Isabella, N. Li, C. B. Kurtz, Developing a new class of engineered live bacterial therapeutics to treat human diseases. *Nat. Commun.* **11**, 1738 (2020). doi: 10.1038/s41467-020-15508-1; pmid: 32269218
152. J. H. Zheng *et al*., Two-step enhanced cancer immunotherapy with engineered *Salmonella typhimurium* secreting heterologous flagellin. *Sci. Transl. Med.* **9**, eaak9537 (2017). doi: 10.1126/scitranslmed.aak9537; pmid: 28179508
153. N. J. Roberts *et al*., Intratumoral injection of *Clostridium novyi*-NT spores induces antitumor responses. *Sci. Transl. Med.* **6**, 249ra111 (2014). doi: 10.1126/scitranslmed.3008982; pmid: 25122639
154. C. A. Swofford, N. Van Dessel, N. S. Forbes, Quorum-sensing *Salmonella* selectively trigger protein expression within tumors. *Proc. Natl. Acad. Sci. U.S.A.* **112**, 3457–3462 (2015). doi: 10.1073/pnas.1414558112; pmid: 25737556
155. M. O. Din *et al*., Synchronized cycles of bacterial lysis for in vivo delivery. *Nature* **536**, 81–85 (2016). doi: 10.1038/nature18930; pmid: 27437587
156. S. Chowdhury *et al*., Programmable bacteria induce durable tumor regression and systemic antitumor immunity. *Nat. Med.* **25**, 1057–1063 (2019). doi: 10.1038/s41591-019-0498-z; pmid: 31270504
157. S. R. Scott *et al*., A stabilized microbial ecosystem of self-limiting bacteria using synthetic quorum-regulated lysis. *Nat. Microbiol.* **2**, 17083 (2017). doi: 10.1038/nmicrobiol.2017.83; pmid: 28604679
158. E. R. Davenport *et al*., The human microbiome in evolution. *BMC Biol.* **15**, 127 (2017). doi: 10.1186/s12915-017-0454-7; pmid: 29282061
159. K. R. Foster, J. Schluter, K. Z. Coyte, S. Rakoff-Nahoum, The evolution of the host microbiome as an ecosystem on a leash. *Nature* **548**, 43–51 (2017). doi: 10.1038/nature23292; pmid: 28770836
160. A. Janney, F. Powrie, E. H. Mann, Host-microbiota maladaptation in colorectal cancer. *Nature* **585**, 509–517 (2020). doi: 10.1038/s41586-020-2729-3; pmid: 32968260
161. R. Z. Gharaibeh, C. Jobin, Microbiota and cancer immunotherapy: In search of microbial signals. *Gut* **68**, 385–388 (2019). doi: 10.1136/gutjnl-2018-317220; pmid: 30530851
162. A. Limeta, B. Ji, M. Levin, F. Gatto, J. Nielsen, Meta-analysis of the gut microbiota in predicting response to cancer immunotherapy in metastatic melanoma. *JCI Insight* **5**, e140940 (2020). doi: 10.1172/jci.insight.140940; pmid: 33268597
163. R. Sinha *et al*., Assessment of variation in microbial community amplicon sequencing by the Microbiome Quality Control (MBQC) project consortium. *Nat. Biotechnol.* **35**, 1077–1086 (2017). doi: 10.1038/nbt.3981; pmid: 28967885
164. R. Eisenhofer *et al*., Contamination in low microbial biomass microbiome studies: Issues and recommendations. *Trends Microbiol.* **27**, 105–117 (2019). doi: 10.1016/j.tim.2018.11.003; pmid: 30497919
165. C. A. Lozupone *et al*., Meta-analyses of studies of the human microbiota. *Genome Res.* **23**, 1704–1714 (2013). doi: 10.1101/gr.151803.112; pmid: 23861384
166. A. Gonzalez *et al*., Qiita: Rapid, web-enabled microbiome meta-analysis. *Nat. Methods* **15**, 796–798 (2018). doi: 10.1038/s41592-018-0141-9; pmid: 30275573
167. N. M. Davis, D. M. Proctor, S. P. Holmes, D. A. Relman, B. J. Callahan, Simple statistical identification and removal of contaminant sequences in marker-gene and metagenomics data. *Microbiome* **6**, 226 (2018). doi: 10.1186/s40168-018-0605-2; pmid: 30558668
168. D. Davar *et al*., Fecal microbiota transplant overcomes resistance to anti–PD-1 therapy in melanoma patients. *Science* **371**, 595–602 (2021). doi: 10.1126/science.abf3363; pmid: 33542131
169. C. R. Merritt *et al*., Multiplex digital spatial profiling of proteins and RNA in fixed tissue. *Nat. Biotechnol.* **38**, 586–599 (2020). doi: 10.1038/s41587-020-0472-9; pmid: 32393914
170. H. Shi *et al*., Highly multiplexed spatial mapping of microbial communities. *Nature* **588**, 676–681 (2020). doi: 10.1038/s41586-020-2983-4; pmid: 33268897
171. S. Min, S. Kim, S.-W. Cho, Gastrointestinal tract modeling using organoids engineered with cellular and microbiota niches. *Exp. Mol. Med.* **52**, 227–237 (2020). doi: 10.1038/s12276-020-0386-0; pmid: 32103122
172. J. Abed *et al*., Fap2 mediates *Fusobacterium nucleatum* colorectal adenocarcinoma enrichment by binding to tumor-expressed Gal-GalNAc. *Cell Host Microbe* **20**, 215–225 (2016). doi: 10.1016/j.chom.2016.07.006; pmid: 27512904
173. J. Abed *et al*., Colon cancer-associated *Fusobacterium nucleatum* may originate from the oral cavity and reach colon tumors via the circulatory system. *Front. Cell. Infect. Microbiol.* **10**, 400 (2020). doi: 10.3389/fcimb.2020.00400; pmid: 32850497
174. T. M. Wassenaar, Insights from 100 years of research with probiotic *E. coli*. *Eur. J. Microbiol. Immunol.* **6**, 147–161 (2016). doi: 10.1556/1886.2016.00029; pmid: 27766164
175. C. R. Gurbatri *et al*., Engineered probiotics for local tumor delivery of checkpoint blockade nanobodies. *Sci. Transl. Med.* **12**, eaax0876 (2020). doi: 10.1126/scitranslmed.aax0876; pmid: 32051224
176. Y. Tanouchi, R. P. Smith, L. You, Engineering microbial systems to explore ecological and evolutionary dynamics. *Curr. Opin. Biotechnol.* **23**, 791–797 (2012). doi: 10.1016/j.copbio.2012.01.006; pmid: 22310174
177. M. J. Liao, M. O. Din, L. Tsimring, J. Hasty, Rock-paper-scissors: Engineered population dynamics increase genetic stability. *Science* **365**, 1045–1049 (2019). doi: 10.1126/science.aaw0542; pmid: 31488693
178. J.-C. Lagier *et al*., Culturing the human microbiota and culturomics. *Nat. Rev. Microbiol.* **16**, 540–550 (2018). doi: 10.1038/s41579-018-0041-0; pmid: 29937540
179. V. Thorsson *et al*., The immune landscape of cancer. *Immunity* **48**, 812–830.e14 (2018). doi: 10.1016/j.immuni.2018.03.023; pmid: 29628290

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Erratum

Post date 26 September 2024

Erratum for the Review "The microbiome and human cancer" by G. D. Sepich-Poore *et al*.

Reference 13 in the Review "The microbiome and human cancer" (26 March 2021, eabc4552) is G. D. Poore *et al*., "Microbiome analyses of blood and tissues suggest cancer diagnostic approach," which was published online 11 March 2020 in *Nature*. *Nature* retracted Poore *et al.* on 26 June 2024 (https://www.nature.com/articles/s41586-024-07656-x), and thus that paper cannot be used to support arguments made in the Review. Figure 3A has been revised to remove citations to reference 13 and elements that relied only on reference 13; additionally, text in the sections "Cancer microbiome diagnostics," "Modulation of the cancer microbiome," and "Outlook for the cancer microbiome" has been amended to remove reliance on reference 13. Reference 13 remains in the reference list but is marked as retracted.

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Gregory D. Sepich-Poore, Laurence Zitvogel, Ravid Straussman, Jeff Hasty, Jennifer A. Wargo, and Rob Knight

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